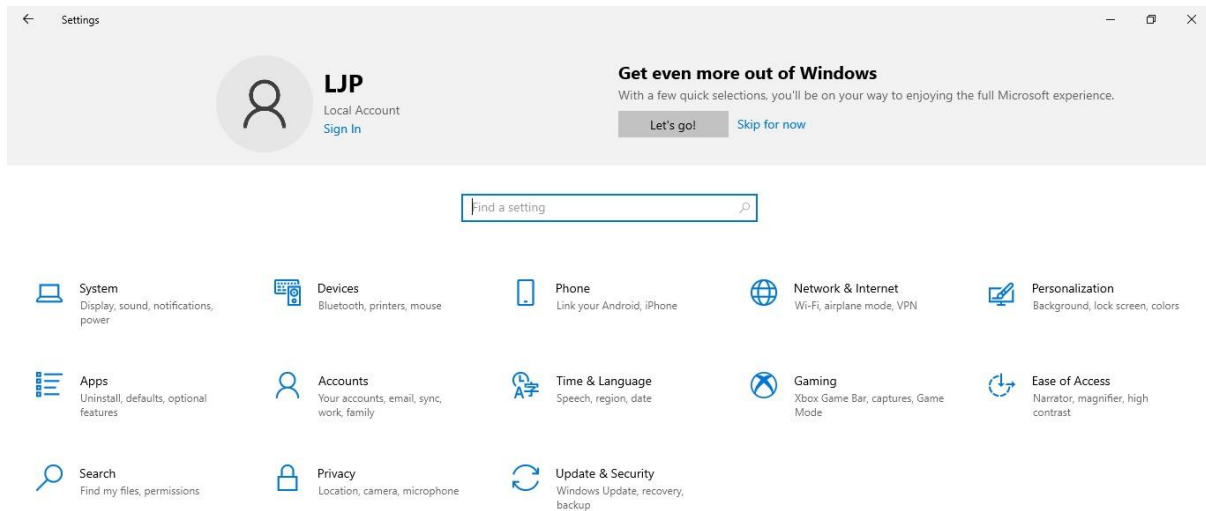


## Experiment: 11

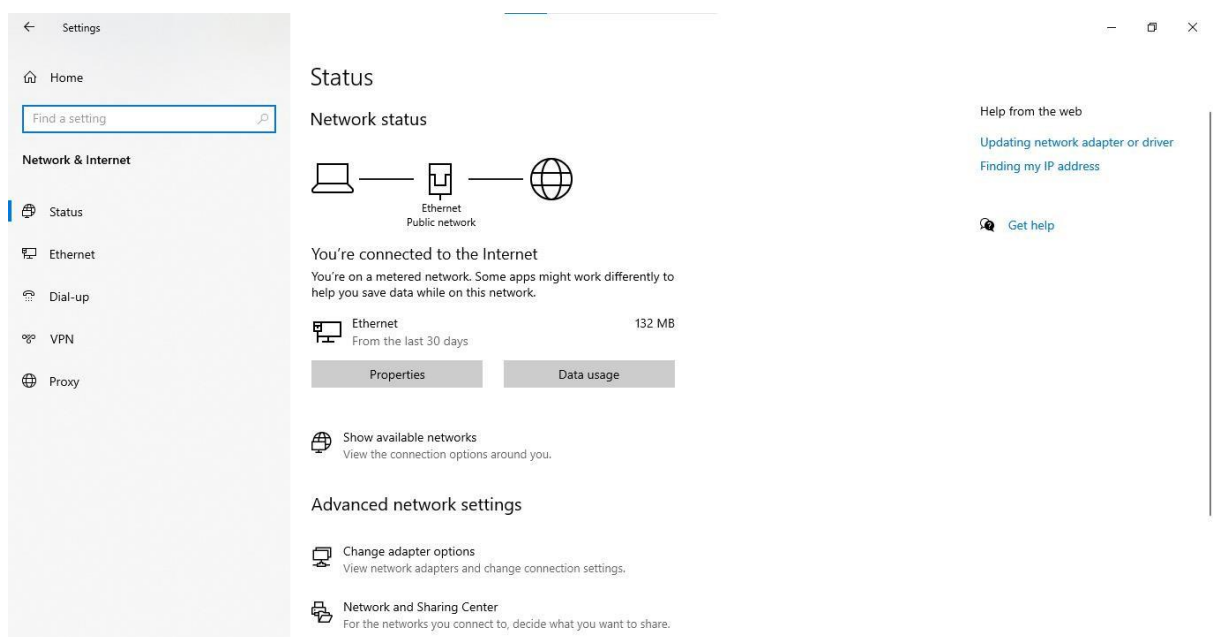
**Aim: Enable DHCP and change TCP/IP settings on the laptop/PC.**

**Step 1: Select Start then select Settings > Network & Internet .**



**Step 2: Do one of the following:**

- For a Wi-Fi network, select Wi-Fi > Manage known networks. Choose the network you want to change the settings for, then select Properties.**
- For an Ethernet network, select Ethernet, then select the Ethernet network you're connected to.**



**Step 3: Under IP assignment, select Edit.**

← Settings

## Unidentified network

### Metered connection

If you have a limited data plan and want more control over data usage, make this connection a metered network. Some apps might work differently to reduce data usage when you're connected to this network.

Set as metered connection



If you set a data limit, Windows will set the metered connection setting for you to help you stay under your limit.

[Set a data limit to help control data usage on this network](#)

### IP settings

IP assignment: Automatic (DHCP)

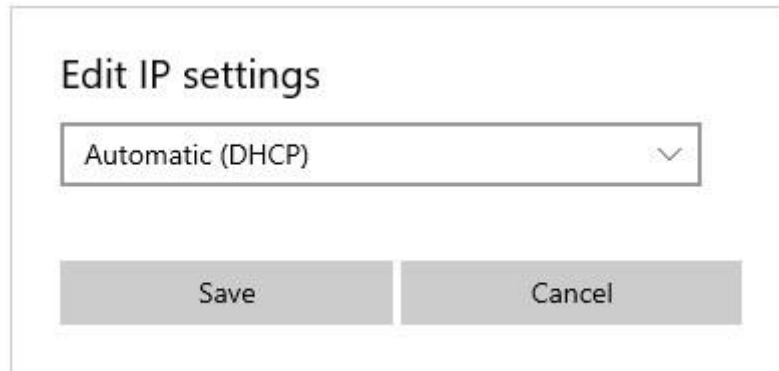
Edit

### Properties

|                                |                              |
|--------------------------------|------------------------------|
| Link speed (Receive/Transmit): | 1000/1000 (Mbps)             |
| Link-local IPv6 address:       | fe80::d7df:8402:69a3:c94d%12 |
| IPv4 address:                  | 192.168.1.178                |

**Step 4: Select Automatic (DHCP) or Manual under Edit IP settings.**

---



The screenshot shows a dialog box titled "Edit IP settings". Inside the dialog, there is a dropdown menu currently displaying "Automatic (DHCP)". Below the dropdown, there are two buttons: "Save" and "Cancel".

**Step 5: To specify IPv4 settings manually, under Edit IP settings, choose Manual, then turn on IPv4.**



The screenshot shows the "Edit IP settings" dialog box. The dropdown menu now displays "Manual". Below the dropdown, there are two sections: "IPv4" and "IPv6". Each section has a toggle switch that is currently in the "Off" position. At the bottom of the dialog, there are "Save" and "Cancel" buttons.

**Step 6: Fill the details of IP address, Subnet Prefix Length, and Gateway. Save the IP address settings.**

---

IPv4

☒ On

IP address

Subnet prefix length

Gateway

×

Preferred DNS

Alternate DNS

IPv6

☐ Off

Save

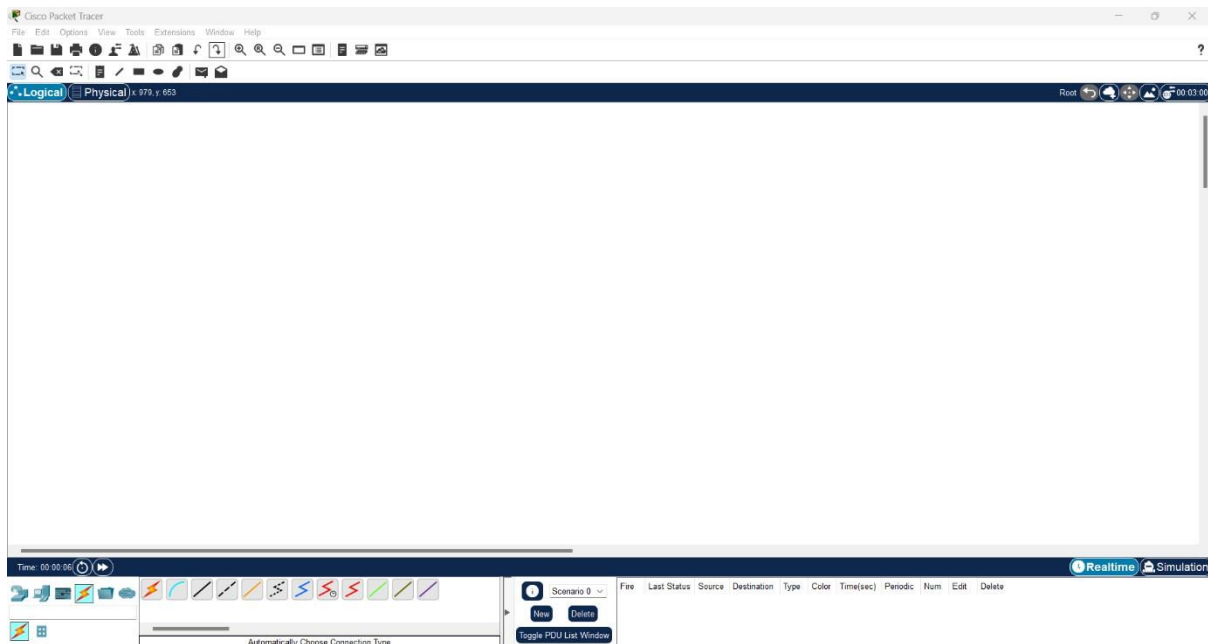
Cancel

## Experiment: 12

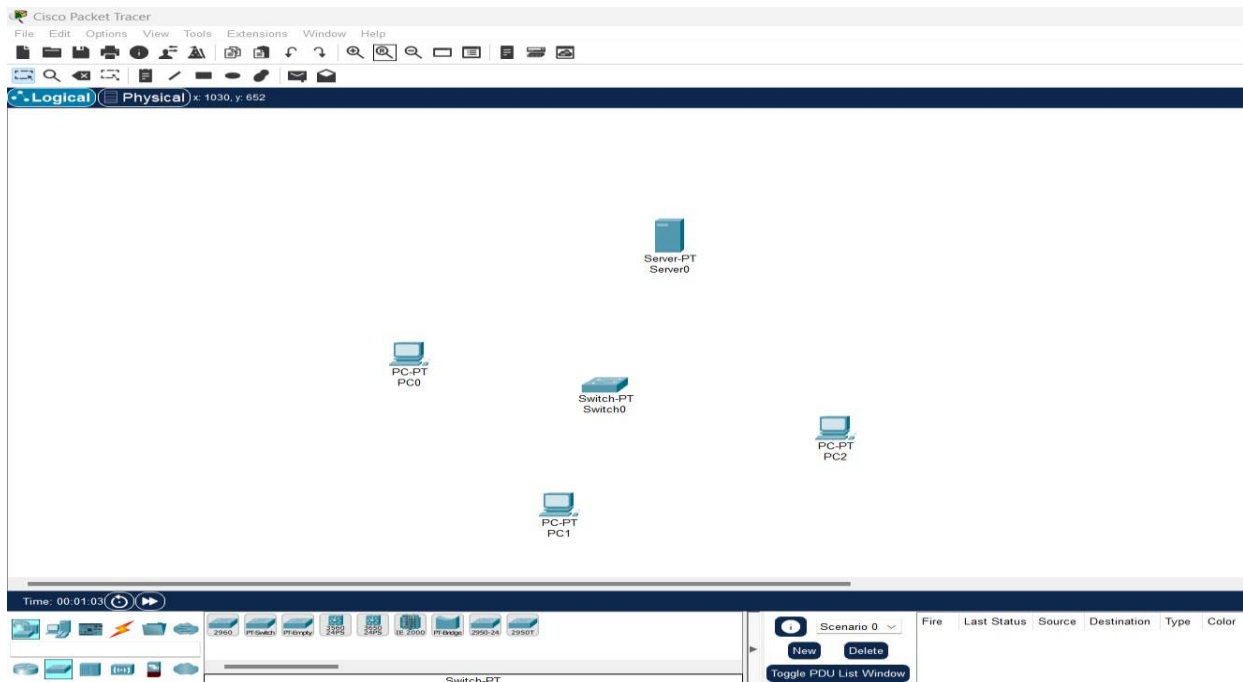
**Aim: Configure DHCP using Cisco Packet Tracer.**

**Step 1: Open Cisco Packet Tracer and create a new project.**

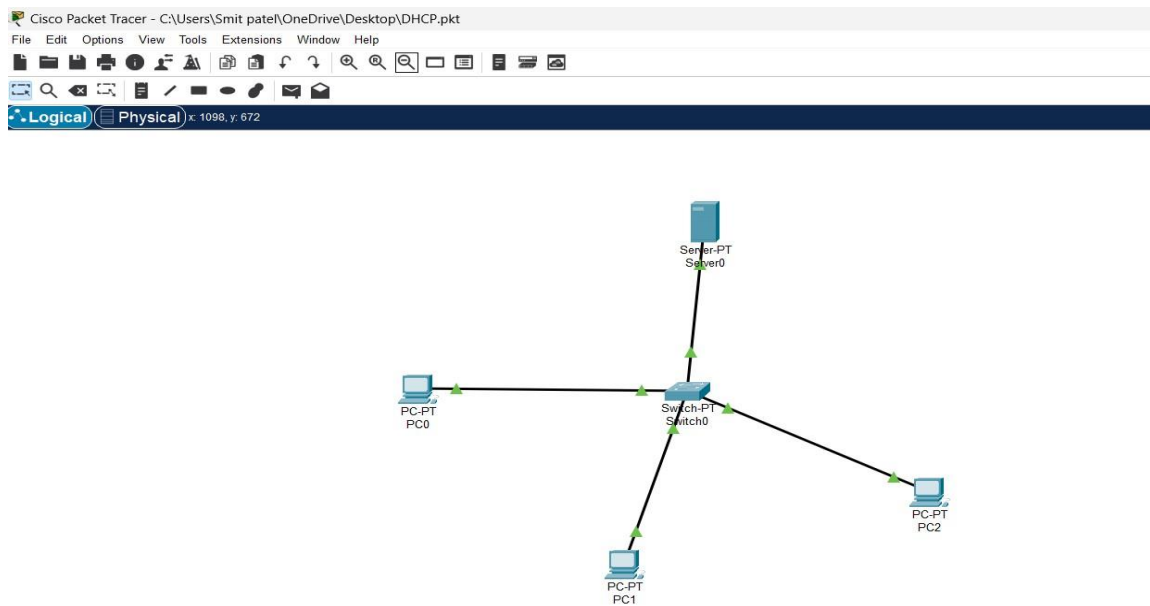
---



**Step 2: Add devices to your workspace. Server** (for the DHCP server), **Switch** (to connect devices), **PCs** (clients that will receive IP addresses from DHCP).



**Step 3: Connect all the devices using appropriate cables. (Choose Automatic Connection Type).**

**Step 4: Configure DHCP Server.**

- To configure the DHCP server first,
- Click on Server then, Go to Services.
- Click on DHCP and turn on the services and, configure the DHCP server with the help of the data given below.
- Delete the default values of the Start IP Address and subnet Mask then save the info.
- Create two new pools.

The screenshot shows the 'Server0' configuration window with the 'Services' tab selected. The 'DHCP' section is active, showing configuration for the 'FastEthernet0' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'gogle..com'. The 'Default Gateway' is '192.168.1.0'. The 'DNS Server' is '0.0.0.0'. The 'Start IP Address' is '192.168.1.2'. The 'Subnet Mask' is '0.0.0.0'. The 'Maximum Number of Users' is '512'. The 'TFTP Server' is '0.0.0.0'. The 'WLC Address' is '0.0.0.0'. Below the configuration fields is a table with columns: Pool Name, Default Gateway, DNS Server, Start IP Address, Subnet Mask, Max User, TFTP Server, and WLC Address. The table contains two rows: 'gogle..com' and 'serverPool'.

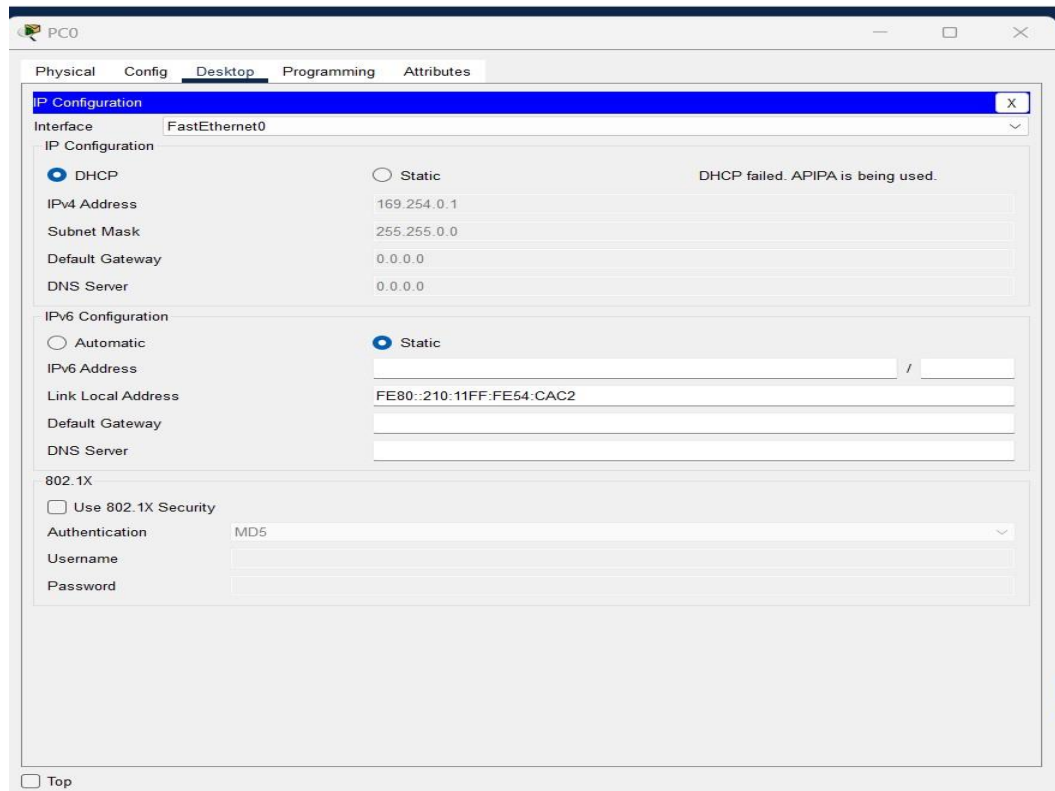
| Pool Name  | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|------------|-----------------|------------|------------------|-------------|----------|-------------|-------------|
| gogle..com | 192.168.1.0     | 0.0.0.0    | 192.168.1.2      | 0.0.0.0     | 512      | 0.0.0.0     | 0.0.0.0     |
| serverPool | 0.0.0.0         | 0.0.0.0    | 0.0.0.0          | 0.0.0.0     | 512      | 0.0.0.0     | 0.0.0.0     |

**Step 5: Select Desktop and fill DHCP field.**

The screenshot shows the 'Server0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active. The 'DHCP' radio button is selected. The 'IPv4 Address' is '169.254.104.216'. The 'Subnet Mask' is '255.255.0.0'. The 'Default Gateway' is '0.0.0.0'. The 'DNS Server' is '0.0.0.0'. The 'IPv6 Configuration' section shows the 'Static' radio button selected. The 'IPv6 Address' is empty. The 'Link Local Address' is 'FE80::20A:F3FF:FE9B:68D8'. The 'Default Gateway' is empty. The 'DNS Server' is empty. The '802.1X' section shows the 'Use 802.1X Security' checkbox unchecked. The 'Authentication' dropdown is set to 'MD5'. The 'Username' and 'Password' fields are empty.

**Step 6: Configure the PCs and change the IP configuration.**

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and IP configuration and there you will find IPv4 configuration.
- Change its state from static to DHCP.
- It will automatically fetch the data and configure itself.
- Repeat the same procedure with other PCs to configure them thoroughly.



PC1

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP failed. APIPA is being used.

IPv4 Address 169.254.0.2

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:96FF:FE0E:49B1

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP failed. APIPA is being used.

IPv4 Address 169.254.0.3

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20C:CFFF:FE85:AC75

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

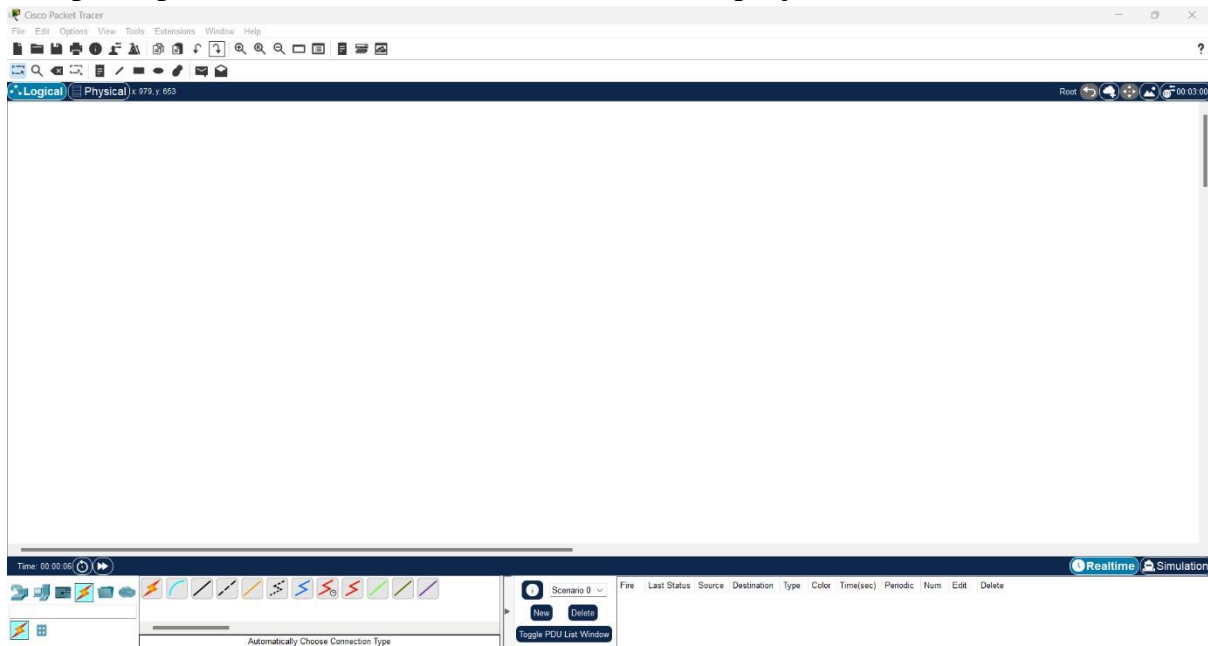
**Step 7: Check the Connection Status.**



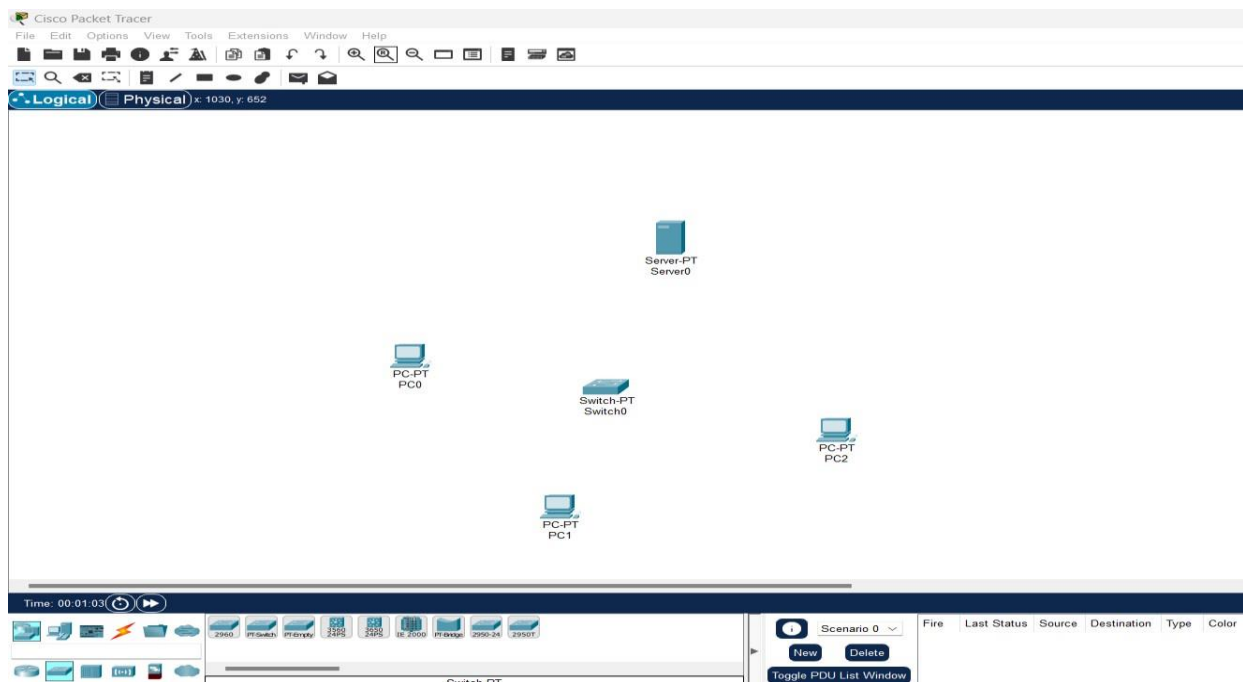
## Experiment: 13

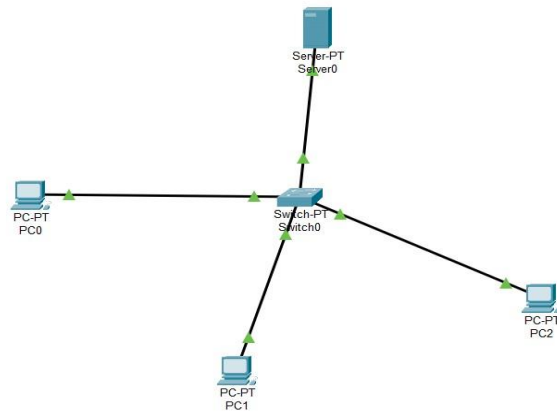
**Aim: Configure DNS Server using Cisco Packet Tracer.**

---

**Step 1: Open Cisco Packet Tracer and create a new project.****Step 2: Add Devices to your workspace:**

- **Switch**, **PCs** (clients that will use the DNS server), **Server** (to act as the DNS server)

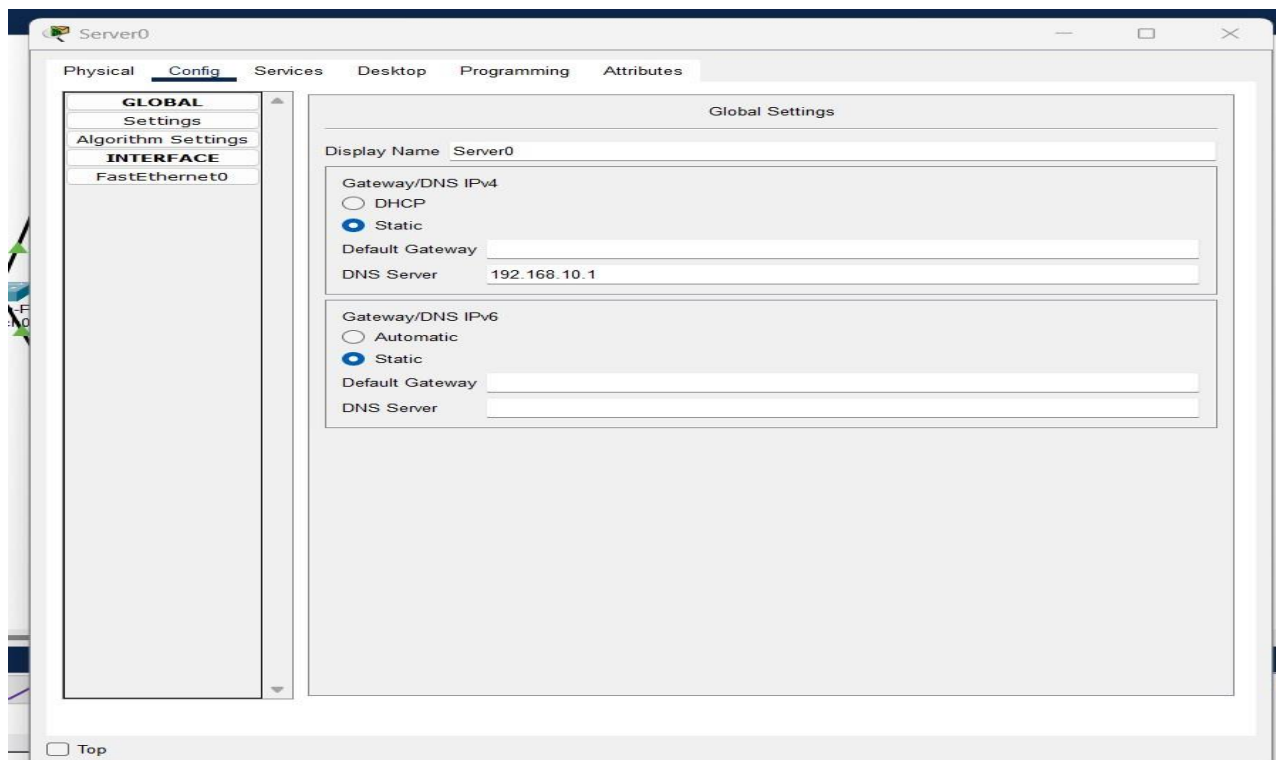
**Step 3: Connect all the devices using the appropriate cables. (Choose Automatic Connection Type).**



#### Step 4: Configure the DNS Server.

- Select the Server and go to its Config tab.
- Configure the IP Address:

- Go to the IPv4 tab and assign an IP address to the server (e.g., 192.168.1.2) with the subnet mask 255.255.255.0 and default gateway as 192.168.1.1.
- **Configure DNS Services:**
  - Click on the DNS tab.
  - Enable DNS service by toggling the switch to “ON”.
  - Add DNS records:
    - **Domain Name:** Enter a domain name you want to resolve (e.g., example.local).
    - **IP Address:** Enter the corresponding IP address of the server or resource (e.g., 192.168.1.3).
  - Add more records as needed (e.g., for multiple domains or subdomains).



Server0

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 192.168.10.1

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20B:BEFF:FE24:A256

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

**DNS**

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name  Type

Address

| No. | Name        | Type     | Detail       |
|-----|-------------|----------|--------------|
| 0   | googles.com | A Record | 192.168.10.0 |
| 1   | googles.com | A Record | 192.168.10.1 |

☐ Top

### Step 5: Configure the PCs

- Select a PC and go to its Desktop tab.
- Click on IP Configuration:
  - Set the PC to use DHCP or assign a static IP address within the same subnet as the server.
  - If using DHCP: Ensure the DHCP configuration on the router provides the DNS server IP address.
  - If using a static IP: Enter the DNS server IP address manually.

The screenshot shows the configuration window for PC0, specifically the Desktop tab. The 'IP Configuration' section is active, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected for both IPv4 and IPv6 configurations. The IPv4 configuration includes an IP address of 192.168.10.2, a subnet mask of 255.255.255.0, a default gateway of 0.0.0.0, and a DNS server of 192.168.10.1. The IPv6 configuration includes a static IP address, a link local address of FE80::209:7CFF:FE6E:8538, and empty fields for default gateway and DNS server. The '802.1X' section is also visible, with 'Use 802.1X Security' unchecked, 'Authentication' set to MD5, and empty fields for 'Username' and 'Password'. A 'Top' button is located at the bottom left of the window.

| IP Configuration                             |                                         |
|----------------------------------------------|-----------------------------------------|
| Interface: FastEthernet0                     |                                         |
| IP Configuration                             |                                         |
| <input type="radio"/> DHCP                   | <input checked="" type="radio"/> Static |
| IPv4 Address                                 | 192.168.10.2                            |
| Subnet Mask                                  | 255.255.255.0                           |
| Default Gateway                              | 0.0.0.0                                 |
| DNS Server                                   | 192.168.10.1                            |
| IPv6 Configuration                           |                                         |
| <input type="radio"/> Automatic              | <input checked="" type="radio"/> Static |
| IPv6 Address                                 |                                         |
| Link Local Address                           | FE80::209:7CFF:FE6E:8538                |
| Default Gateway                              |                                         |
| DNS Server                                   |                                         |
| 802.1X                                       |                                         |
| <input type="checkbox"/> Use 802.1X Security |                                         |
| Authentication                               | MD5                                     |
| Username                                     |                                         |
| Password                                     |                                         |

☐ Top

The screenshot shows the 'PC2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'Static' option is selected for both IPv4 and IPv6 configurations. The IPv4 configuration includes an address of 192.168.10.4, a subnet mask of 255.255.255.0, a default gateway of 0.0.0.0, and a DNS server of 192.168.10.1. The IPv6 configuration is also set to 'Static' with a link local address of FE80::201:63FF:FE72:A2A2. The '802.1X' section is collapsed.

| Interface                                                               | FastEthernet0            |
|-------------------------------------------------------------------------|--------------------------|
| IP Configuration                                                        |                          |
| <input type="radio"/> DHCP <input checked="" type="radio"/> Static      |                          |
| IPv4 Address                                                            | 192.168.10.4             |
| Subnet Mask                                                             | 255.255.255.0            |
| Default Gateway                                                         | 0.0.0.0                  |
| DNS Server                                                              | 192.168.10.1             |
| IPv6 Configuration                                                      |                          |
| <input type="radio"/> Automatic <input checked="" type="radio"/> Static |                          |
| IPv6 Address                                                            |                          |
| Link Local Address                                                      | FE80::201:63FF:FE72:A2A2 |
| Default Gateway                                                         |                          |
| DNS Server                                                              |                          |
| 802.1X                                                                  |                          |
| <input type="checkbox"/> Use 802.1X Security                            |                          |
| Authentication                                                          | MD5                      |
| Username                                                                |                          |
| Password                                                                |                          |

### Step 6: Check Connection Status.

The screenshot shows the 'Realtime Simulation' window with a table of network events. The table has columns for Fire, Last Status, Source, Destination, Type, Color, Time(sec), Periodic, Num, Edit, and Delete. There are three rows of data, all showing 'Successful' status and 'ICMP' type.

| Fire | Last Status | Source  | Destination | Type | Color | Time(sec) | Periodic | Num | Edit   | Delete   |
|------|-------------|---------|-------------|------|-------|-----------|----------|-----|--------|----------|
|      | Successful  | Server0 | PC0         | ICMP |       | 0.000     | N        | 0   | (edit) | (delete) |
|      | Successful  | Server0 | PC1         | ICMP |       | 0.000     | N        | 1   | (edit) | (delete) |
|      | Successful  | Server0 | PC2         | ICMP |       | 0.000     | N        | 2   | (edit) | (delete) |

## Experiment: 14

**Aim: Execute Basic TCP/IP utilities and commands. (Eg: ping, ipconfig, tracert, arp, host, netstat, nslookup, ftp, telnet, etc).**

### 1) Ping

Ping is the primary TCP/IP command used to troubleshoot connectivity, reachability, and name resolution. Used without parameters, this command displays Help content. You can also use this command to test both the computer name and the IP address of the computer.

```
C:\Users\LJP>ping 192.168.45.153

Pinging 192.168.45.153 with 32 bytes of data:
Reply from 192.168.45.153: bytes=32 time<1ms TTL=128
Reply from 192.168.45.153: bytes=32 time<1ms TTL=128
Reply from 192.168.45.153: bytes=32 time<1ms TTL=128
Reply from 192.168.45.153: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.45.153:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\LJP>
```

**2) ipconfig** ipconfig ("Internet Protocol configuration") is a console application program of some computer operating systems that displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings.

---

Ipcfg is a command-line utility that displays and manages a computer's network settings and IP address

```

C:\Users\gespl>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::506d:8173:76d7:170e%48
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 11:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 12:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 2:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2409:40c1:25:e3f8:8d67:f166:464b:bb36
    Temporary IPv6 Address. . . . . : 2409:40c1:25:e3f8:a4a3:f0ed:aa82:9707
    Link-local IPv6 Address . . . . . : fe80::8d67:f166:464b:bb36%11
    IPv4 Address. . . . . : 172.20.10.3
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : fe80::8820:dff:fe24:7c64%11
                                172.20.10.1

```

### 3) Tracert

The tracert command is Windows' version of traceroute. It is used for tracing the path between two systems. It will display the different routers or hops needed to travel from the source system to the destination system. Tracert will also show the time it takes to get from the source to the destination.

---

```
C:\Users\LJP>tracert
```

```
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
           [-R] [-S srcaddr] [-4] [-6] target_name
```

```
Options:
```

```
-d          Do not resolve addresses to hostnames.
-h maximum_hops  Maximum number of hops to search for target.
-j host-list  Loose source route along host-list (IPv4-only).
-w timeout    Wait timeout milliseconds for each reply.
-R          Trace round-trip path (IPv6-only).
-S srcaddr    Source address to use (IPv6-only).
-4          Force using IPv4.
-6          Force using IPv6.
```

```
C:\Windows\System32>tracert google.com
```

```
Tracing route to google.com [2404:6800:4002:81c::200e]
over a maximum of 30 hops:
```

|    |       |       |       |                                                      |
|----|-------|-------|-------|------------------------------------------------------|
| 1  | *     | *     | *     | Request timed out.                                   |
| 2  | 5 ms  | 3 ms  | 3 ms  | undefined.hostname.localhost [2402:a00:404:1::1111]  |
| 3  | 5 ms  | 4 ms  | 4 ms  | undefined.hostname.localhost [2402:a00:1::3d]        |
| 4  | 45 ms | 18 ms | 18 ms | undefined.hostname.localhost [2402:a00::2e]          |
| 5  | 48 ms | 23 ms | 23 ms | 2001:4860:1:1::2818                                  |
| 6  | 18 ms | 18 ms | 18 ms | 2404:6800:812f::1                                    |
| 7  | 23 ms | 18 ms | 18 ms | 2001:4860:0:1::539a                                  |
| 8  | 22 ms | 18 ms | 18 ms | 2001:4860:0:1::78bc                                  |
| 9  | 19 ms | 20 ms | 19 ms | 2001:4860:0:1::77ad                                  |
| 10 | 21 ms | 18 ms | 19 ms | 2001:4860:0:1::5501                                  |
| 11 | 17 ms | 19 ms | 20 ms | del11s17-in-x0e.1e100.net [2404:6800:4002:81c::200e] |

```
Trace complete.
```

```
C:\Windows\System32>
```

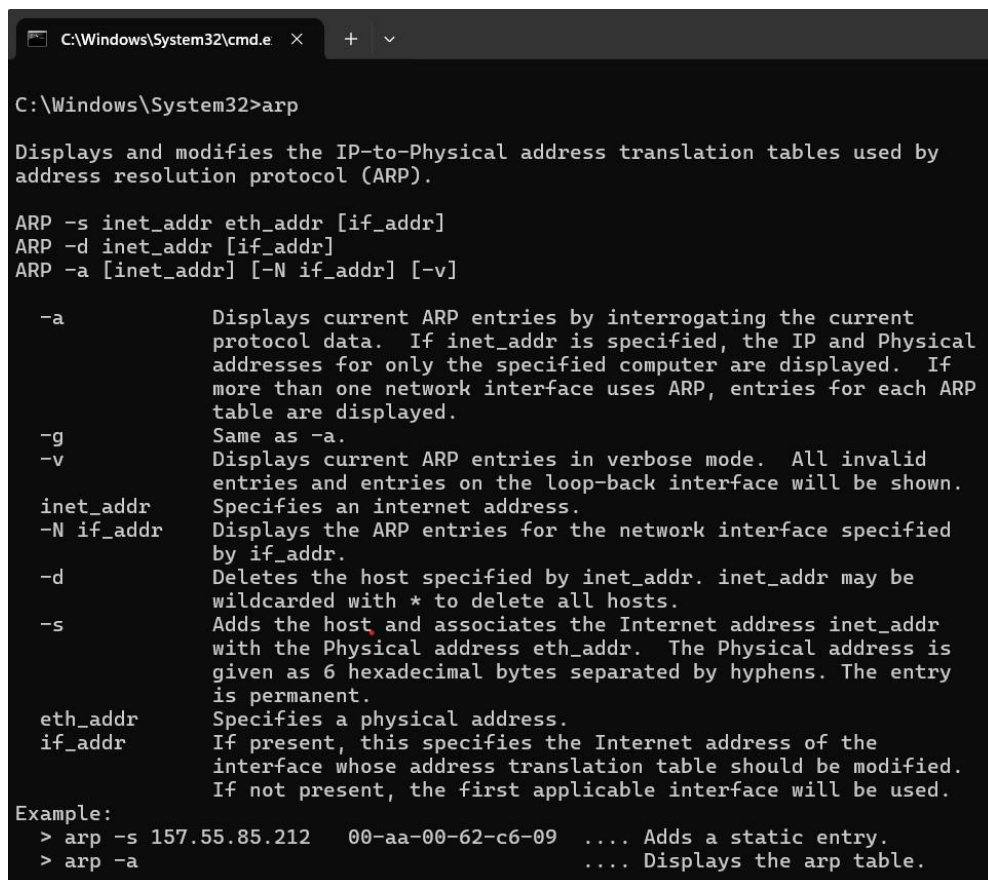
#### 4) arp

The arp command in a computer network displays and modifies the Address Resolution Protocol (ARP) cache. ARP is a process that connects a dynamic IP address to a physical machine's MAC address. This is important because IP and MAC addresses are different lengths, so a translation is needed for systems to recognize each other.

---

The arp command can display the current ARP entry for a host specified by name or number. For example, to access the cache, you can type arp -a to see a list of internet addresses, physical addresses, and the type of entry in the ARP cache. There are two types of entries: dynamic and static. Dynamic entries are used for everyday things and are consumed within two minutes, so you'll need to find them again. Static entries last for the duration of the machine being up and running.

You can also modify the ARP cache using the arp command, but you'll need to have an administrative user account or know the username and password of an administrator account.



```
C:\Windows\System32\cmd.e x + v

C:\Windows\System32>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

-a          Displays current ARP entries by interrogating the current
            protocol data. If inet_addr is specified, the IP and Physical
            addresses for only the specified computer are displayed. If
            more than one network interface uses ARP, entries for each ARP
            table are displayed.
-g          Same as -a.
-v          Displays current ARP entries in verbose mode. All invalid
            entries and entries on the loop-back interface will be shown.
inet_addr   Specifies an internet address.
-N if_addr  Displays the ARP entries for the network interface specified
            by if_addr.
-d          Deletes the host specified by inet_addr. inet_addr may be
            wildcarded with * to delete all hosts.
-s          Adds the host and associates the Internet address inet_addr
            with the Physical address eth_addr. The Physical address is
            given as 6 hexadecimal bytes separated by hyphens. The entry
            is permanent.
eth_addr    Specifies a physical address.
if_addr     If present, this specifies the Internet address of the
            interface whose address translation table should be modified.
            If not present, the first applicable interface will be used.

Example:
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.
> arp -a          .... Displays the arp table.
```

```
C:\Windows\System32>arp -a
```

```
Interface: 192.168.56.1 --- 0x5
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 192.168.1.18 --- 0xc
  Internet Address      Physical Address      Type
  192.168.1.1           b8-dd-71-ee-e1-b8     dynamic
  192.168.1.10          00-46-b8-11-1c-94     dynamic
  192.168.1.113         00-46-b8-11-1f-11     dynamic
  192.168.1.118         00-46-b8-11-39-88     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

### 5) netstat

The netstat command (short for "network statistics") is a command-line utility used to display network connections, routing tables, interface statistics, masquerade connections, and multicast

---

memberships. It's available on various operating systems, including Windows, Linux, and macOS.

Here's a breakdown of the common uses and options for the netstat command:

#### Common Uses

1. Display Active Connections: Shows all active network connections and listening ports.
2. Display Routing Tables: Provides the routing tables used by the system.
3. Display Network Interfaces: Lists network interfaces and their statistics.

```
C:\Users\LJP>netstat -a

Active Connections

Proto Local Address           Foreign Address         State
TCP   0.0.0.0:80              304-88:0                LISTENING
TCP   0.0.0.0:135             304-88:0                LISTENING
TCP   0.0.0.0:445             304-88:0                LISTENING
TCP   0.0.0.0:796             304-88:0                LISTENING
TCP   0.0.0.0:5040            304-88:0                LISTENING
TCP   0.0.0.0:49664           304-88:0                LISTENING
TCP   0.0.0.0:49665           304-88:0                LISTENING
TCP   0.0.0.0:49666           304-88:0                LISTENING
TCP   0.0.0.0:49667           304-88:0                LISTENING
TCP   0.0.0.0:49668           304-88:0                LISTENING
TCP   0.0.0.0:49693           304-88:0                LISTENING
TCP   0.0.0.0:49695           304-88:0                LISTENING
TCP   127.0.0.1:49684          304-88:49685            ESTABLISHED
TCP   127.0.0.1:49685          304-88:49684            ESTABLISHED
TCP   127.0.0.1:49741          304-88:49742            ESTABLISHED
TCP   127.0.0.1:49742          304-88:49741            ESTABLISHED
TCP   192.168.44.88:139        304-88:0                LISTENING
TCP   192.168.44.88:49808      LJP-SERVERROOM:7074     ESTABLISHED
TCP   192.168.44.88:50109      DESKTOP-2FBLRKB:808     TIME_WAIT
TCP   192.168.44.88:50111      DESKTOP-2FBLRKB:808     CLOSE_WAIT
TCP   192.168.44.88:50112      DESKTOP-2FBLRKB:808     CLOSE_WAIT
TCP   192.168.44.88:50114      DESKTOP-2FBLRKB:808     CLOSE_WAIT
```

## 6) nslookup

The nslookup command in Windows is a network utility used for querying DNS (Domain Name System) servers to obtain information about domain names and IP addresses. It helps diagnose

---

DNS-related issues by allowing you to look up domain name information, verify DNS configurations, and check the resolution of hostnames.

```
C:\Windows\System32>nslookup google.com
Server:  gpon.net
Address:  192.168.1.1

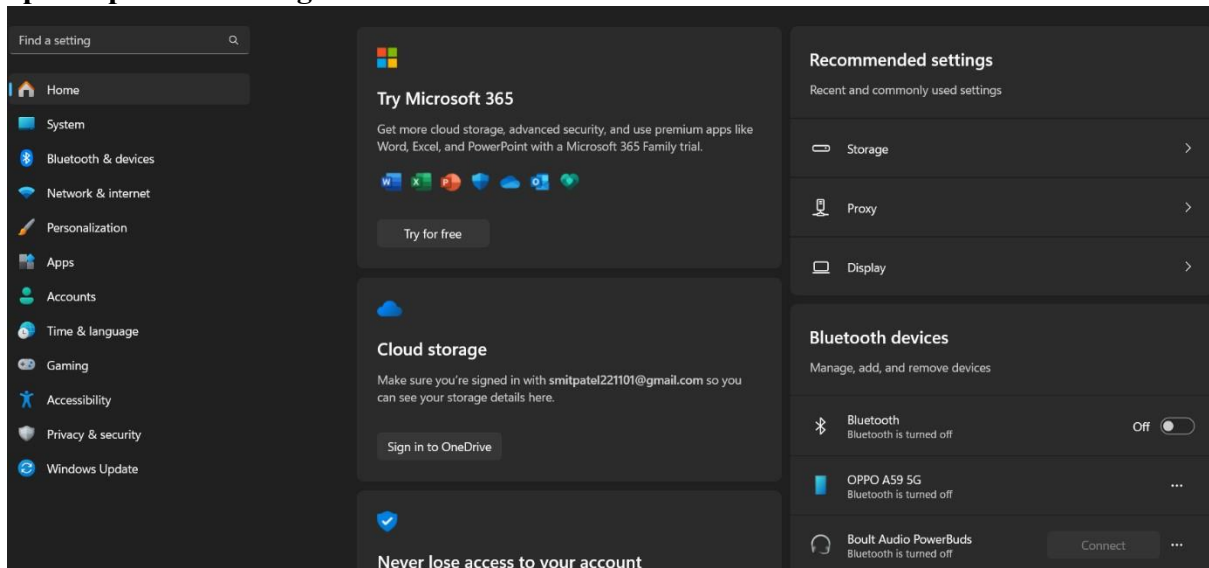
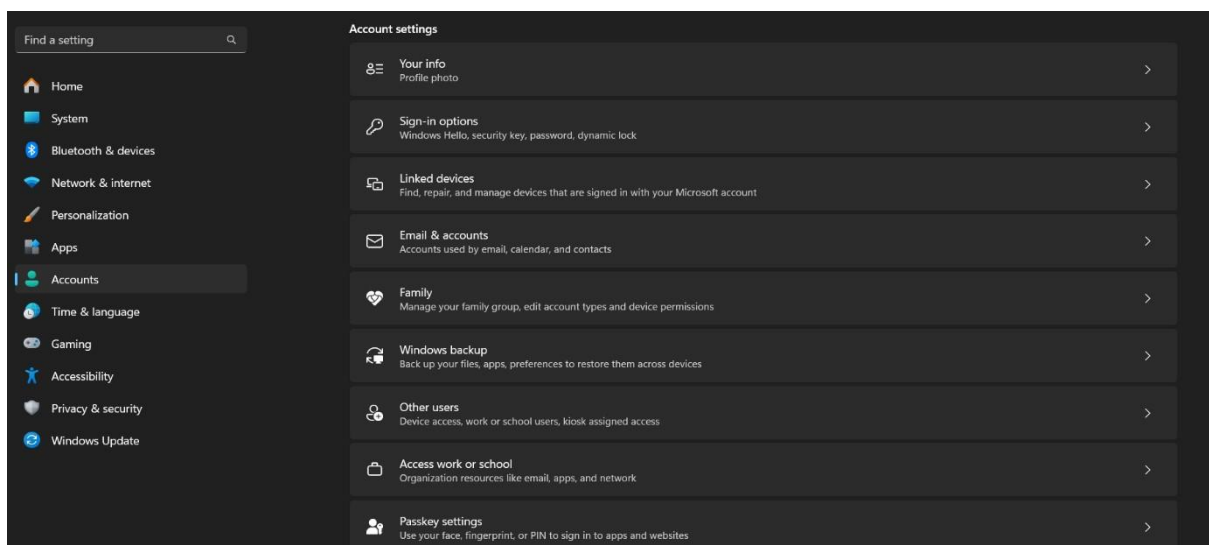
Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4002:81c::200e
          142.250.66.14

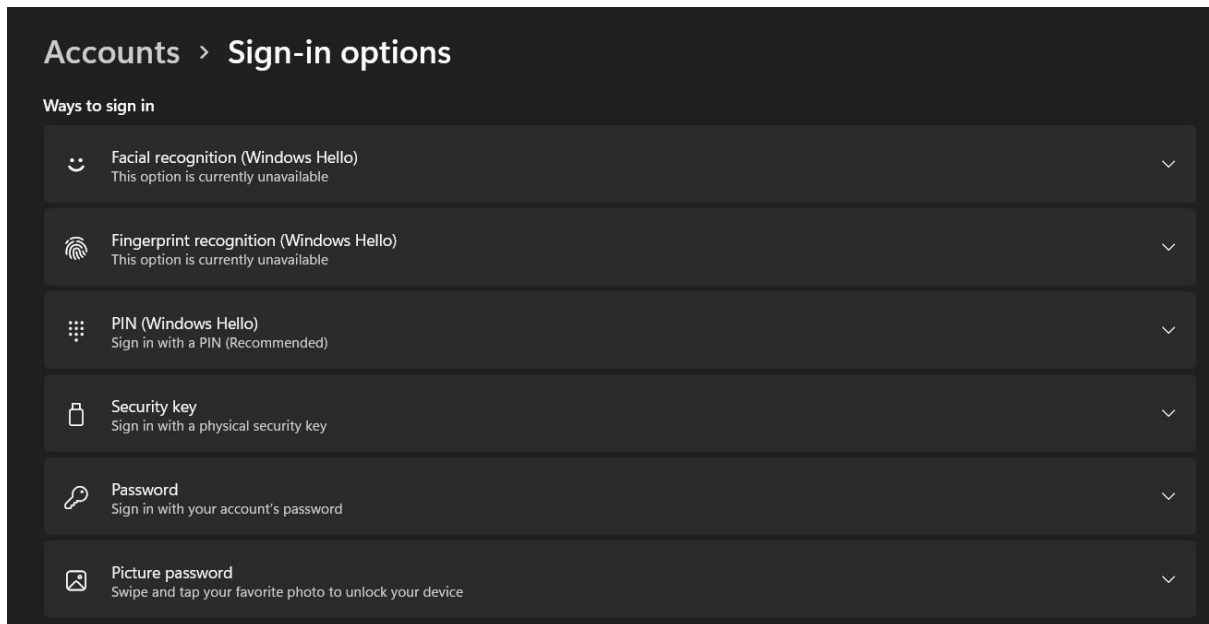
C:\Windows\System32>
```

## **Experiment: 15**

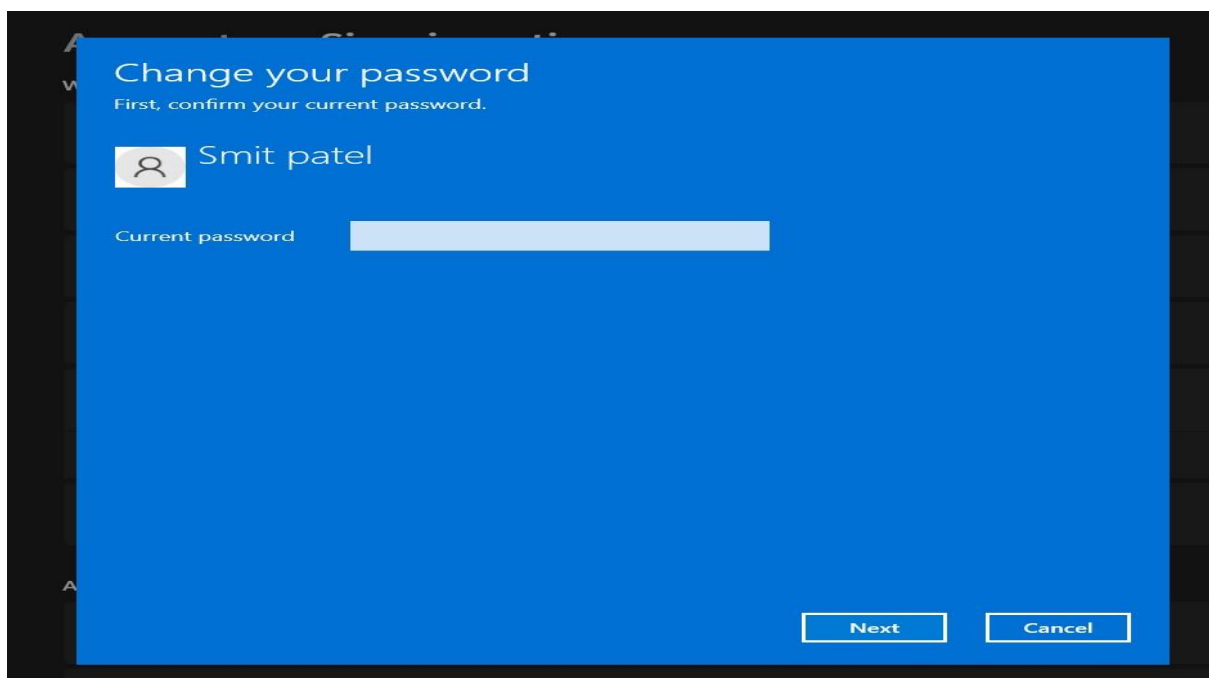
**Aim:** Configure a system for various security experiments like:

- a) Set up a password for every user account.
  - b) Make some standard users and restrict other users from accessing the new users.
-

**a) Set up a password for every user account.****Step 1: Open the Settings.****Step 2: Open User Accounts.****Step 3: Select Sign-in option.**

**Step 4: Change the Password.**

- Click Change the password.
- Enter the current password (if applicable), then enter and confirm the new password.
- Click Change Password to apply.

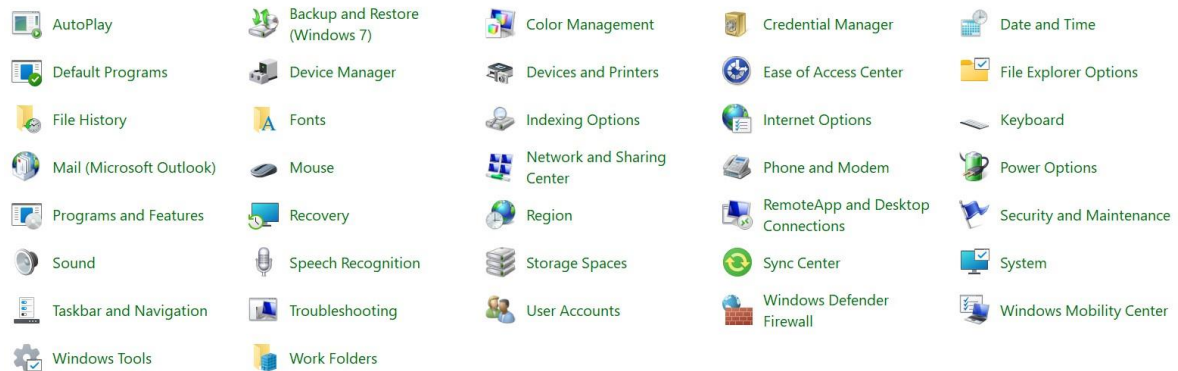
**Step 5: Repeat for other accounts.**

**b) Make some standard users and restrict other users from accessing the new users.**

---

## Step 1: Open Control Panel

Adjust your computer's settings



## Step 2: Click User Account

Control Panel Home

Manage your credentials

Create a password reset disk

Configure advanced user profile properties

Change my environment variables

### Make changes to your user account

Make changes to my account in PC settings

Change your account name

Change your account type

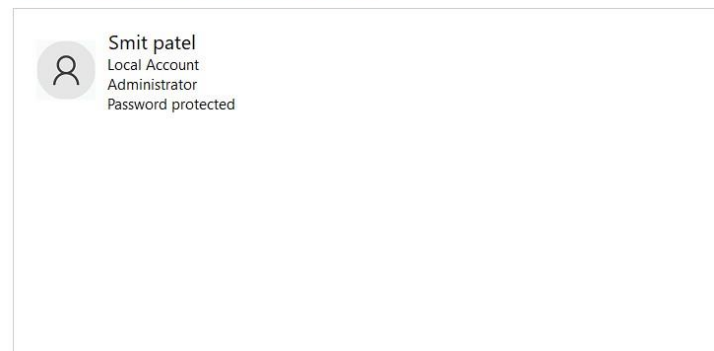
Manage another account

Change User Account Control settings



## Step 3: Click Manage Another account

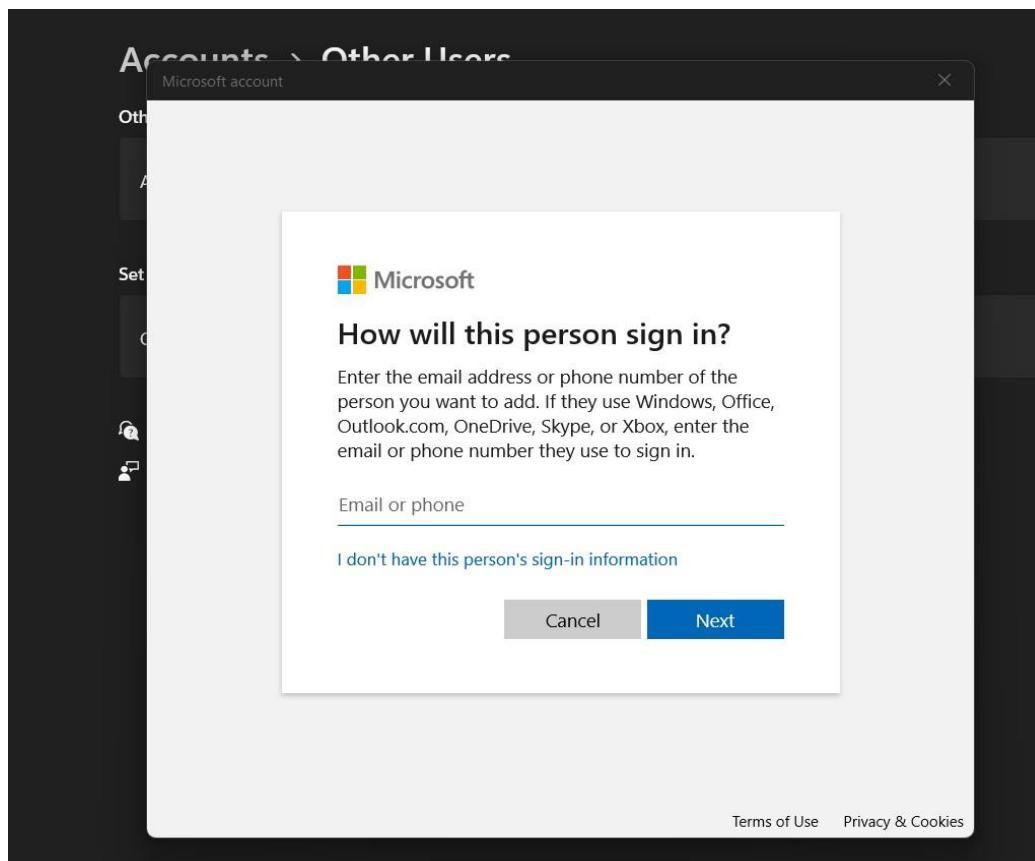
Choose the user you would like to change

A screenshot of a Windows user selection window. It features a single user card for 'Smit patel'. The card includes a generic user icon, the name 'Smit patel', and the roles 'Local Account', 'Administrator', and 'Password protected'.

[Add a new user in PC settings](#)

#### Step 4: Manage a new account.

Enter the name for the new account, select Standard user, and click Create Account.



#### Step 5: Restrict Access to New Users:

- Open Control Panel > User Accounts > Manage another account.

- Select the user account you want to restrict.
- Click Change the account type and ensure it is set to Standard user or Guest.
- Optionally, you can use Local Security Policy (secpol.msc) to configure specific restrictions:
- Open secpol.msc from the Run dialog.
- Navigate to Local Policies > User Rights Assignment.
- Modify settings like Deny log on locally or Deny log on through Remote Desktop Services to restrict access.

## **Experiment: 16**

**Aim: Configure Web browser security settings for:**

- a) Perform various security settings on Google Chrome.**
  - b) Set up a double password on the account's synced data, and set up the browsing to a secured condition.**
-

### **a) Performing various security settings on Google Chrome.**

#### **1) Keep Chrome Updated**

- Updates often include security fixes that protect you from the latest threats.
  1. Click the three dots in the top right corner.
  2. Go to **Help > About Google Chrome**.
  3. Chrome will automatically check for updates. Click **Relaunch** if an update is available.

#### **2) Enable Safe Browsing**

- Safe Browsing protects you from dangerous websites, downloads, and extensions.
  1. Click the three dots in the top right corner.
  2. Go to **Settings > Privacy and Security > Safe Browsing**.
  3. Choose **Enhanced Protection** for better security.

#### **3) Block Third-Party Cookies**

- Blocking third-party cookies helps prevent tracking and improves privacy.
  1. Go to **Settings > Privacy and Security > Cookies and other site data**.
  2. Select **Block third-party cookies**.

#### **4) Use a Secure DNS**

- Secure DNS helps protect against phishing and other threats using a trusted DNS provider.
  1. Go to **Settings > Privacy and Security > Security**.
  2. Select a provider like Google (Public DNS) or Cloudflare under Use Secure DNS.

#### **5) Manage Permissions**

- Control what sites can access (e.g., your location, camera, microphone) to enhance privacy.
    1. Go to **Settings > Privacy and Security > Site Settings**.
    2. Review and adjust permissions for **Location, Camera, Microphone, Notifications**, and other settings.
-

## 6) Enable Site Isolation

- Site Isolation improves security by separating different sites into different processes.
  1. Type `chrome://flags/#enable-site-per-process` in the address bar.
  2. Enable the **Strict Site Isolation** option.
  3. Relaunch Chrome.

## 7) Use Password Manager

- Securely store and manage your passwords.
  1. Go to **Settings > Autofill > Passwords**.
  2. Ensure **Offer to save passwords** and **Auto Sign-in** are enabled.

## 8) Enable 2-Step Verification (2FA)

- Adds an extra layer of security to your Google account.
  1. Visit your Google Account security page at [myaccount.google.com/security](https://myaccount.google.com/security).
  2. Scroll to **2-Step Verification** and click **Get Started**.

## 9) Check for Harmful Software

- Chrome can scan your device for malware that could compromise your security.
  1. Go to **Settings > Reset and clean up > Clean up computer**.
  2. Click **Find** to start the scan.

## 10) Clear Browsing Data Regularly

- Regularly clearing your browsing data helps maintain privacy and security.
  1. Go to **Settings > Privacy and Security > Clear browsing data**.
  2. Choose a time range and the types of data you want to clear.

## **b) Set up a double password on the account's synced data, and set up the browsing to a secured condition.**

### 1) Set Up Double Password Protection on Synced Data

#### A) Enable Sync and Encryption with a Passphrase

- By default, Chrome syncs your data across devices using your Google account. Adding a sync passphrase provides an extra layer of encryption, ensuring that only you can decrypt your synced data.
-

1. Open Chrome and click the three dots in the top right corner.
2. Go to **Settings > You and Google > Sync and Google services**.
3. Under **Sync**, click **Encryption options**.
4. Select **Encrypt synced data with your sync passphrase**.
5. Enter and confirm your passphrase, then click **Save**.
6. Remember this passphrase, as you will need it to access your synced data on any device.

### **B) Set Up Two-Step Verification (2SV) for Your Google Account**

- Two-step verification (2SV) adds an extra layer of security to your Google account, requiring a second factor (like a phone or security key) in addition to your password.
  1. Go to your Google Account security page: [myaccount.google.com/security](https://myaccount.google.com/security).
  2. Scroll to **2-Step Verification** and click **Get Started**.
  3. Follow the prompts to set up 2SV using your phone number, Google Authenticator app, or a security key.

## **2) Set Up Secure Browsing Conditions**

### **A) Enable Safe Browsing with Enhanced Protection**

- Enhanced Safe Browsing offers better protection against phishing, malware, and other web-based threats.
  1. Go to **Settings > Privacy and Security > Safe Browsing**.
  2. Select **Enhanced protection**.

### **B) Manage Site Permissions Strictly**

- Limiting site permissions (e.g., access to your camera, microphone, location) minimizes potential security risks.
  1. Go to **Settings > Privacy and Security > Site Settings**.
  2. Review and adjust permissions for each category (e.g., Location, Camera, Microphone, Notifications).
  3. Set permissions to **Ask before accessing** where possible.

### **C) Block Third-Party Cookies**

---

- Blocking third-party cookies helps prevent websites from tracking your browsing activity.
  1. Go to **Settings > Privacy and Security > Cookies and other site data**.
  2. Select **Block third-party cookies**.

#### **D) Use a Secure DNS**

- Secure DNS helps protect against phishing and other threats using a trusted DNS provider.
  1. Go to **Settings > Privacy and Security > Security**.
  2. Select a provider like Google (Public DNS) or Cloudflare under Use Secure DNS.

#### **E) Enable Site Isolation**

- Site Isolation separates each site's processes, adding a layer of security to prevent crosssite attacks.
  1. Type **chrome://flags/#enable-site-per-process** in the address bar.
  2. Enable the **Strict Site Isolation** option.
  3. Relaunch Chrome.

#### **F) Regularly Clear Browsing Data**

- Regularly clearing browsing data helps maintain privacy by removing cookies, cached images, and browsing history.
  1. Go to **Settings > Privacy and Security > Clear browsing data**.
  2. Choose a time range and the types of data you want to clear, then click **Clear data**.

### **Experiment: 17**

**Aim: Set up and configure local print devices.**

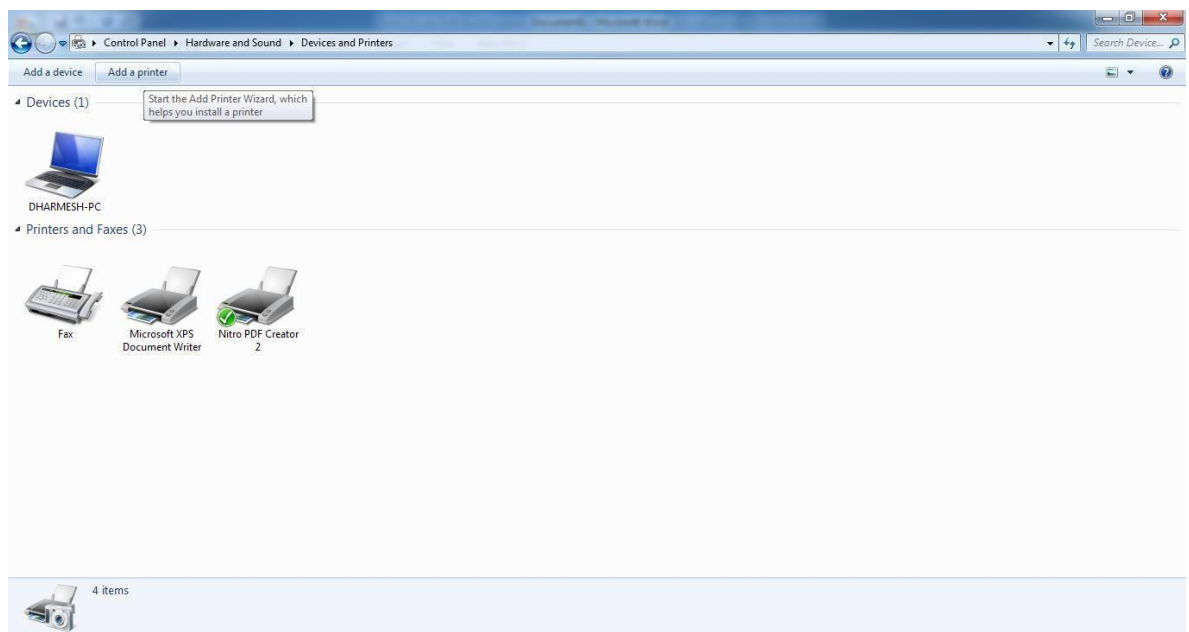
**Step 1: Go to the control panel.**

**Step 2: Click on view devices and printers.**

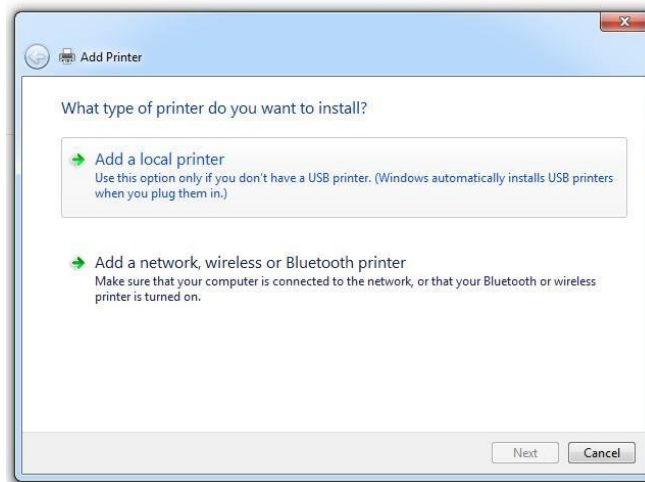
---



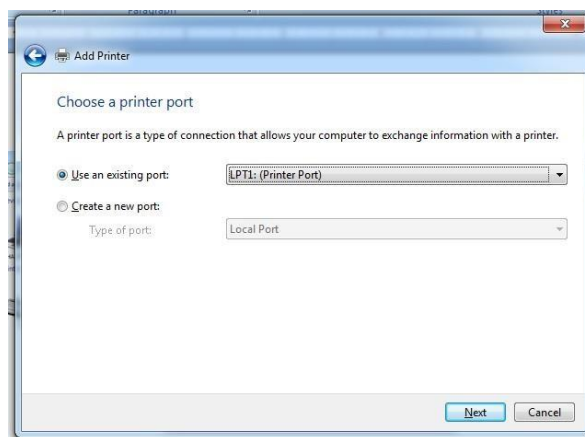
**Step 3: Click on add a printer.**



**Step 4: Choose Add a local printer.**



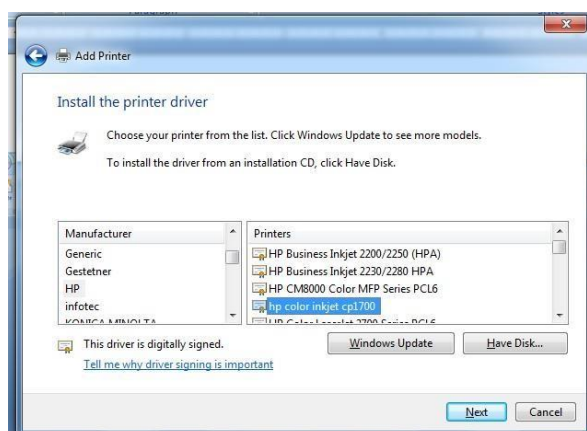
**Step 5: Choose a port for the printer. Select Use an existing port→LPT1→Next.**



**Step 6: Select the printer Manufacturer in the left pane.**

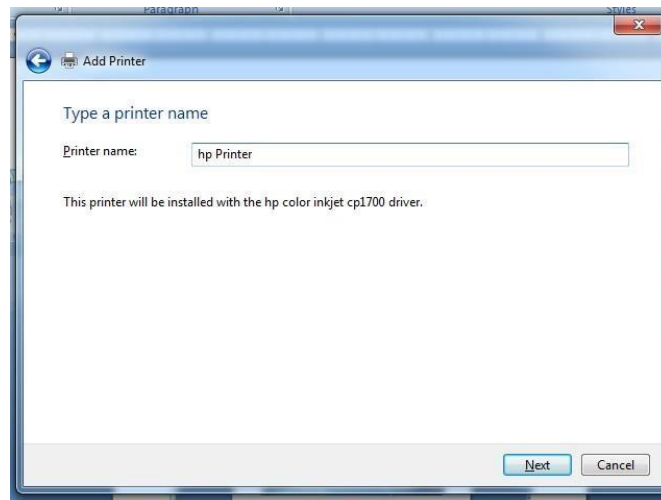
**Step 7: All the printers of that manufacturer will be displayed in the right pane.**

**Select the appropriate printer→ Next.**

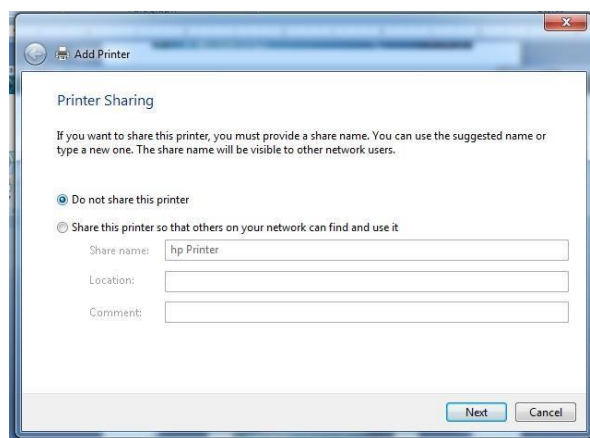


**Step 8: Give a name to the printer→ Next.**

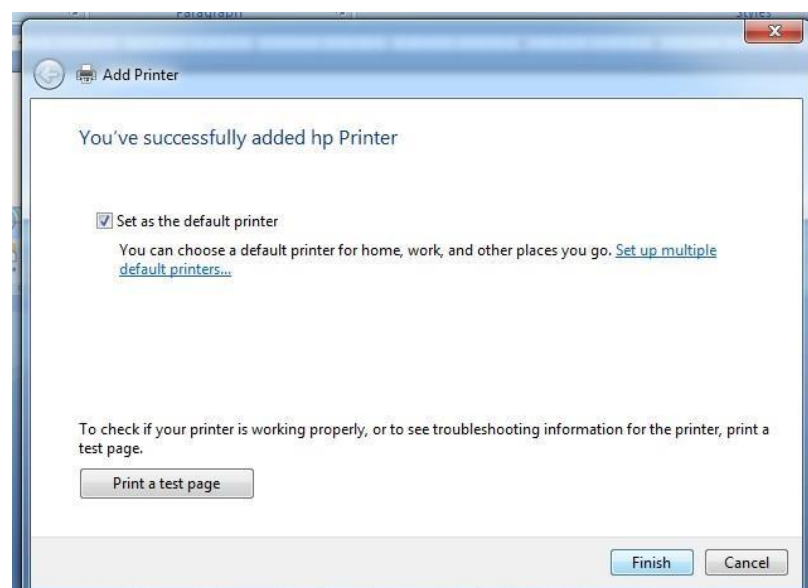
---



**Step 9: In Printer Sharing select → Do not share this printer→ Next.**

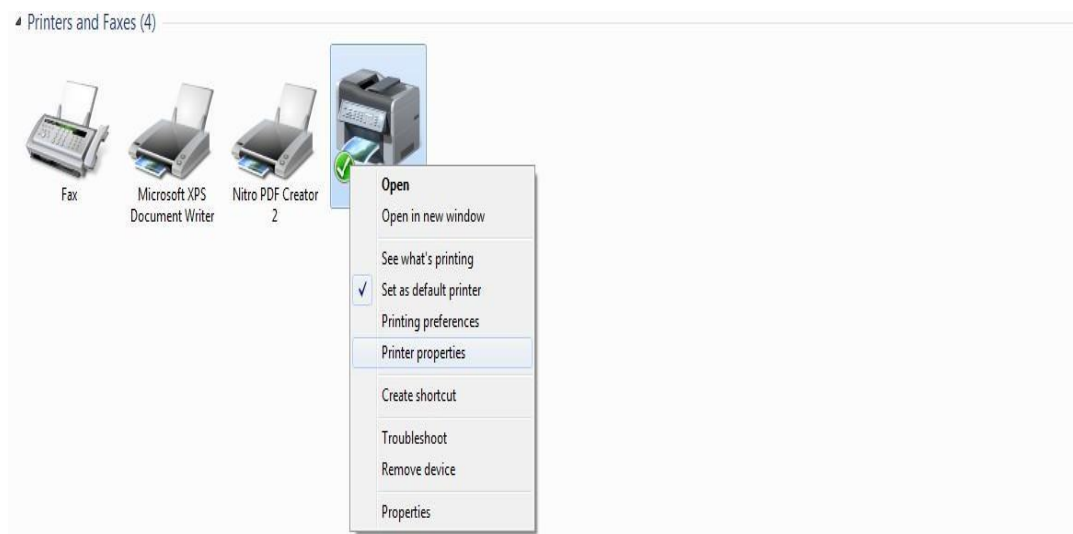


**Step 10: If you want to set it as default then select set as default printer. Otherwise do not select it. Then click on Finish.**

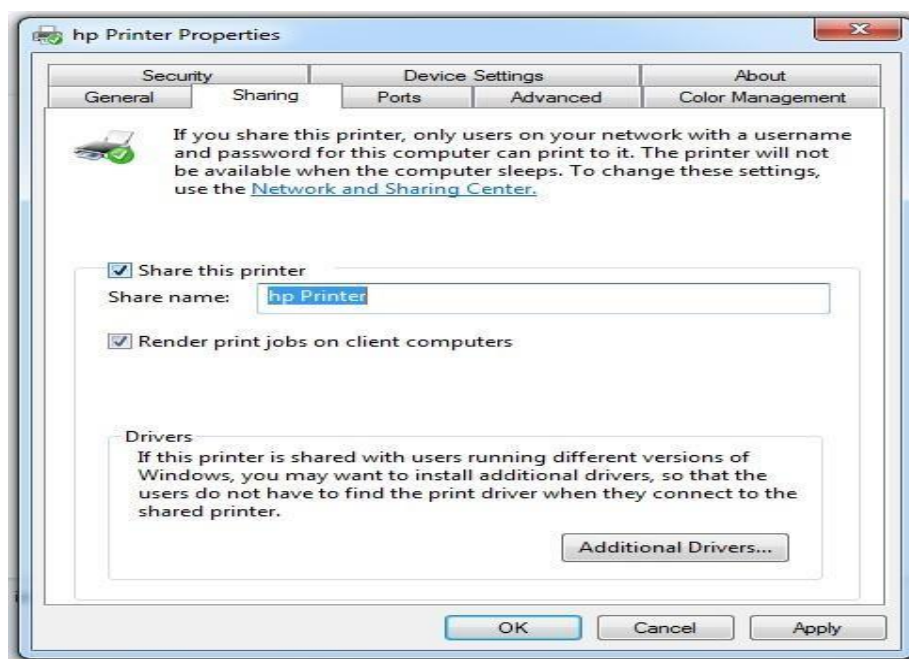


### **Sharing the local Printer**

**Step 1: Go to the control panel then select “view devices and printers”. Right-click the Printer and choose Printer Properties.**



**Step 2: Go to the sharing tab and enable the option “Share this printer”. You can set its Share name. Now click on Apply and then OK.**



## **Experiment: 18 Aim:**

### **Install and Configure Virtual Box.**

#### **Introduction to Virtual Box:**

- VirtualBox is a tool that creates virtual machines (VMs).
- A virtual machine is like a computer inside your computer.
- You can run different operating systems (like Windows, Linux, or macOS) on these virtual machines.

#### **How Does It Work?**

- Installation: You install VirtualBox on your computer, just like any other program.
- Creating a VM: Inside VirtualBox, you create a new virtual machine. You decide how much of your computer's resources (like RAM, CPU, and storage) the VM will use.
- Installing an OS: You install an operating system on the VM just like you would on a real computer. For example, you could install Windows on a VM running on a Mac.
- Running the VM: You start the VM, and it runs the installed operating system in a window on your desktop. You can interact with it as if it were a separate computer.

#### **Why Use VirtualBox?**

- Testing and Development: Developers use VMs to test software in different operating systems without needing multiple computers.
  - Learning: You can experiment with different operating systems and software without risking your main computer.
  - Isolation: VMs can keep activities isolated. For instance, if you suspect a program might be harmful, running it in a VM prevents it from affecting your main system.
-

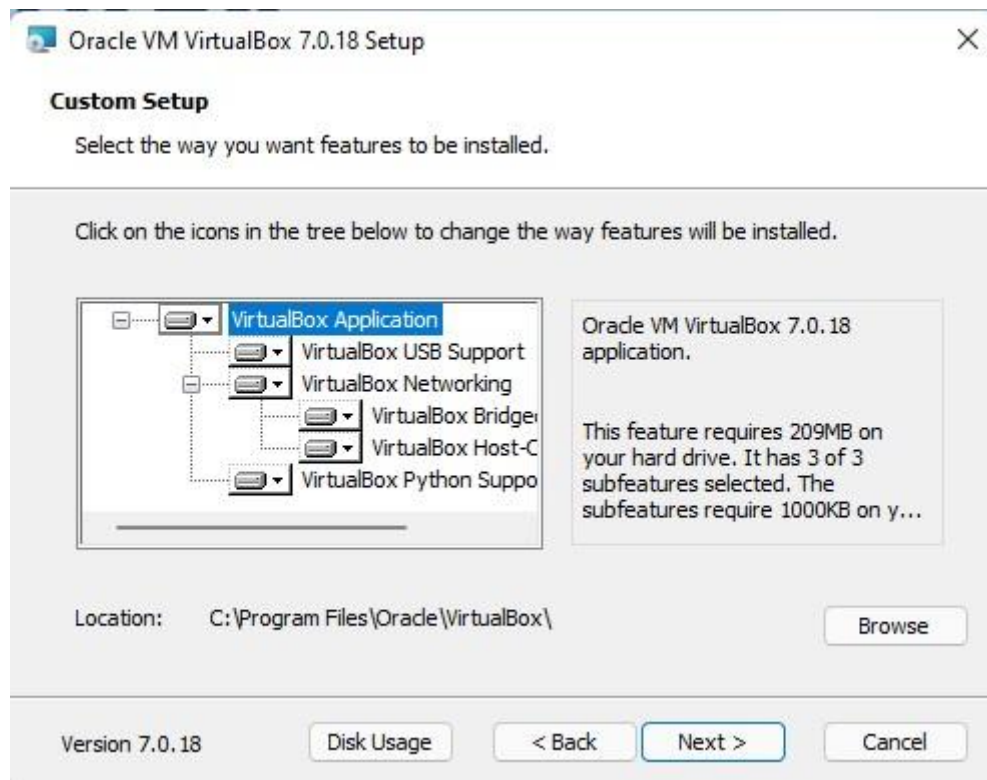
**Step 1: To download VirtualBox, go to the official site <https://www.virtualbox.org/> and download the latest version for Windows.**



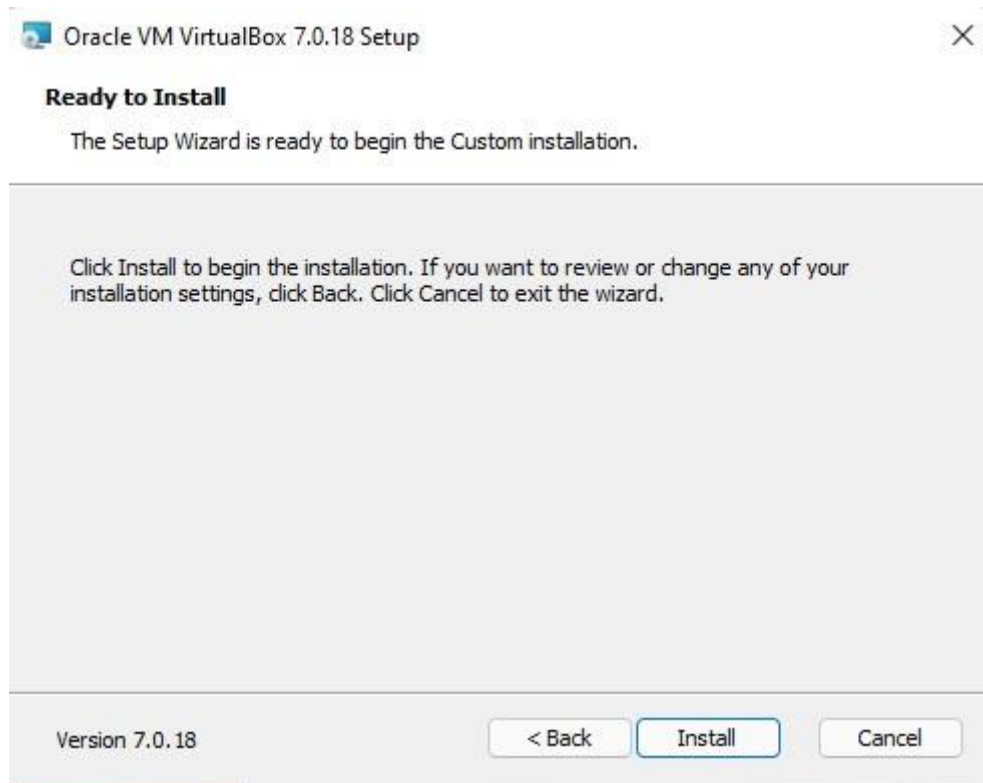
**Step 2: Getting Started**



**Step 3: Select Installation Location**



#### Step 4: Ready to Install



**Step 5: Install Files and packages and after that installation is finished.**



**When you open VirtualBox it will look like this as shown below:**

---



## Experiment: 19

**Aim: Install and Configure Windows 7 or Windows 10 Operating System on Virtual Box.**

**Step 1: Download Virtual Box and Windows ISO.**

- For Windows 10, you can download the ISO directly from Microsoft's official site.
- For Windows 7, you might need to use a valid product key to download the ISO from the Microsoft website or use a trusted source if you have an older copy.

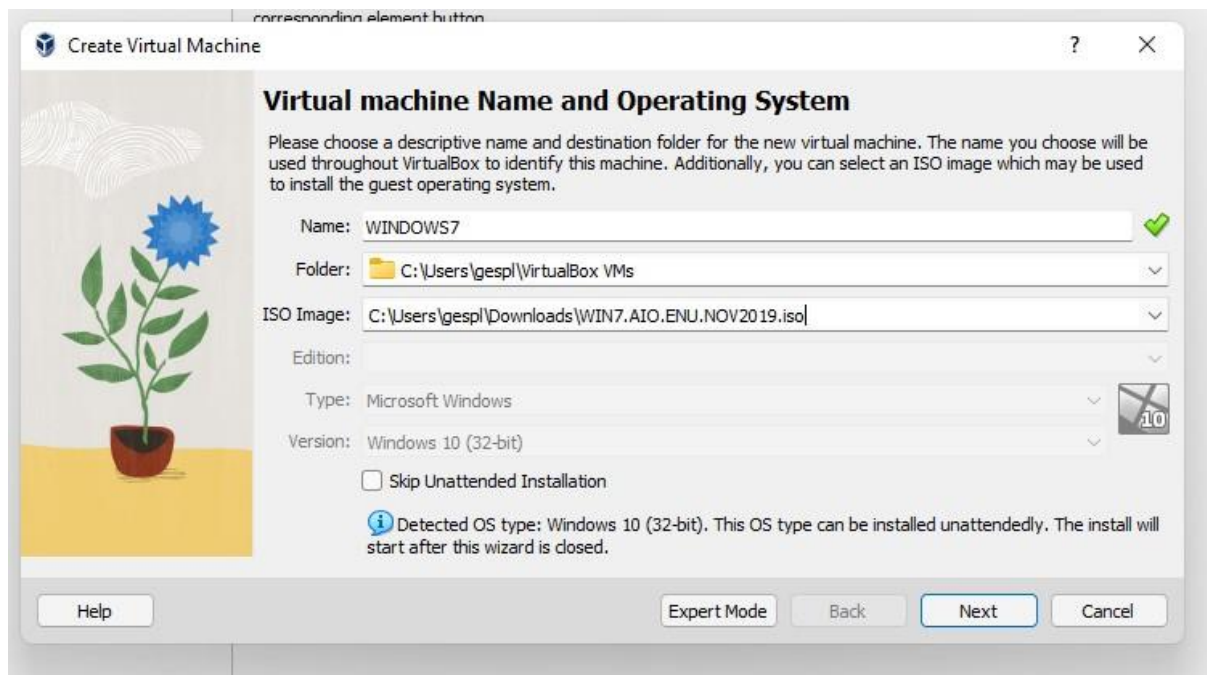
**Step 2: Install Virtual Box.**

**Step 3: Create a new Virtual Machine. In the Machine Section, click on the New Button.**



**Step 4: Enter a name for the new Virtual Machine and select the location of the Windows 7 .iso file.**

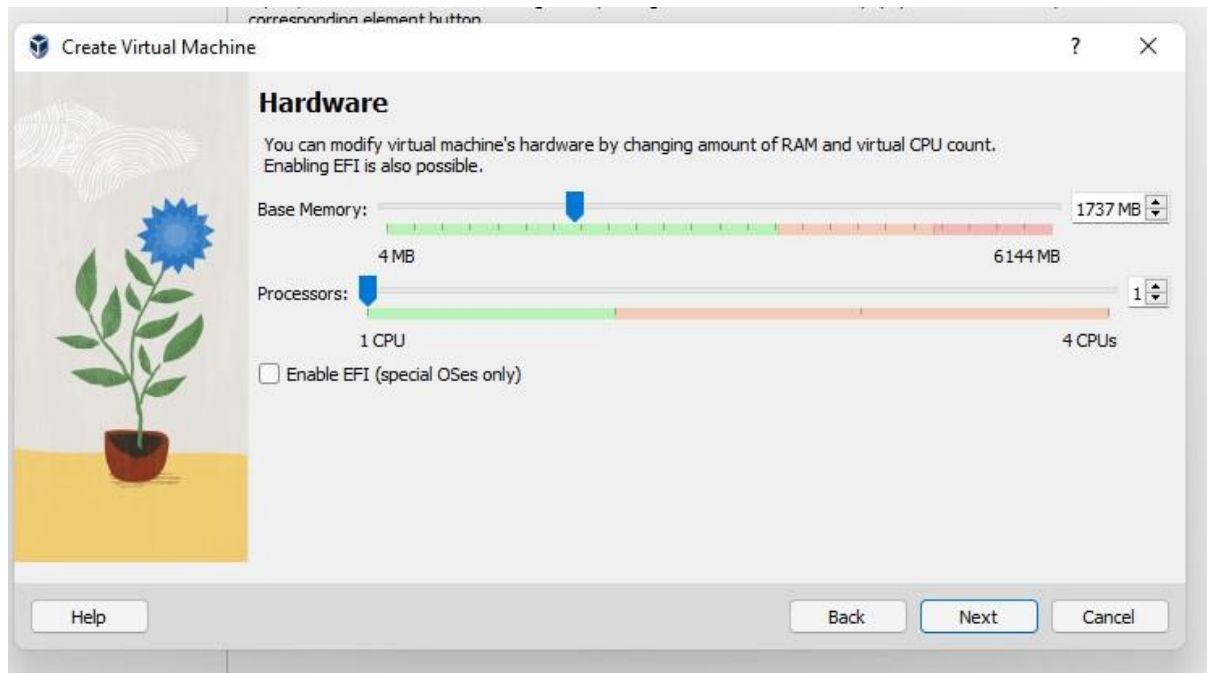
**Click Next.**



**Step 5: Fill up various details as shown in figure below.**



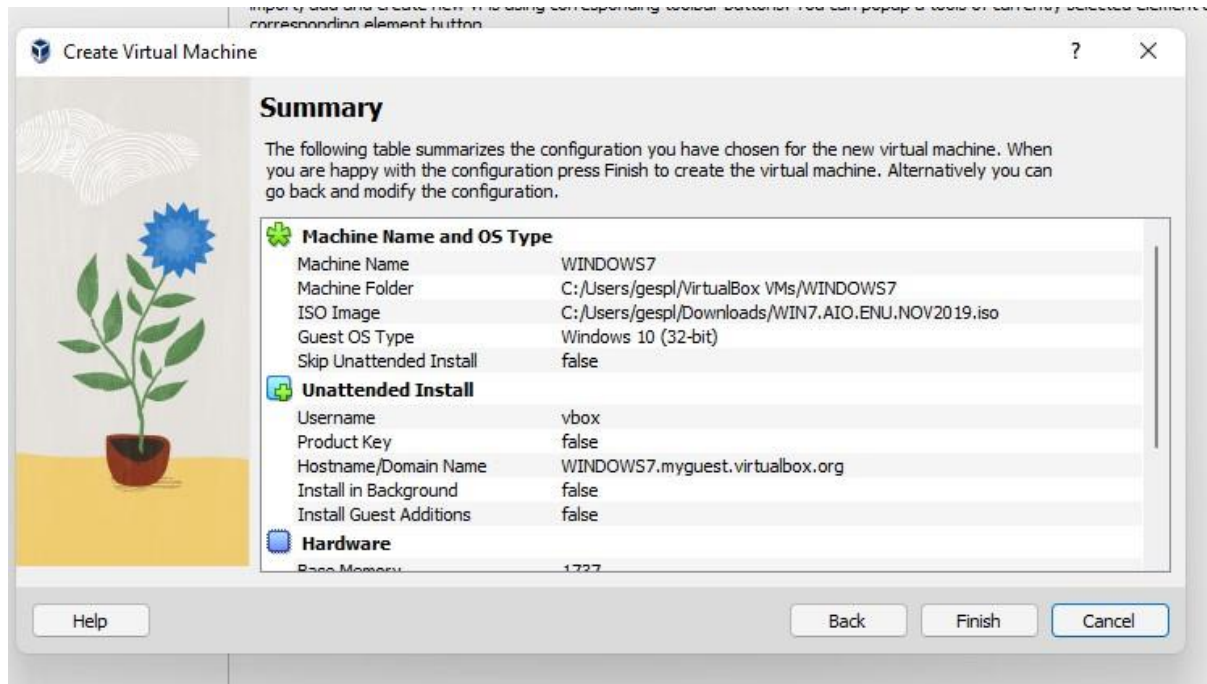
**Step 6: Allocate at least 2 GB of RAM and click Next.**



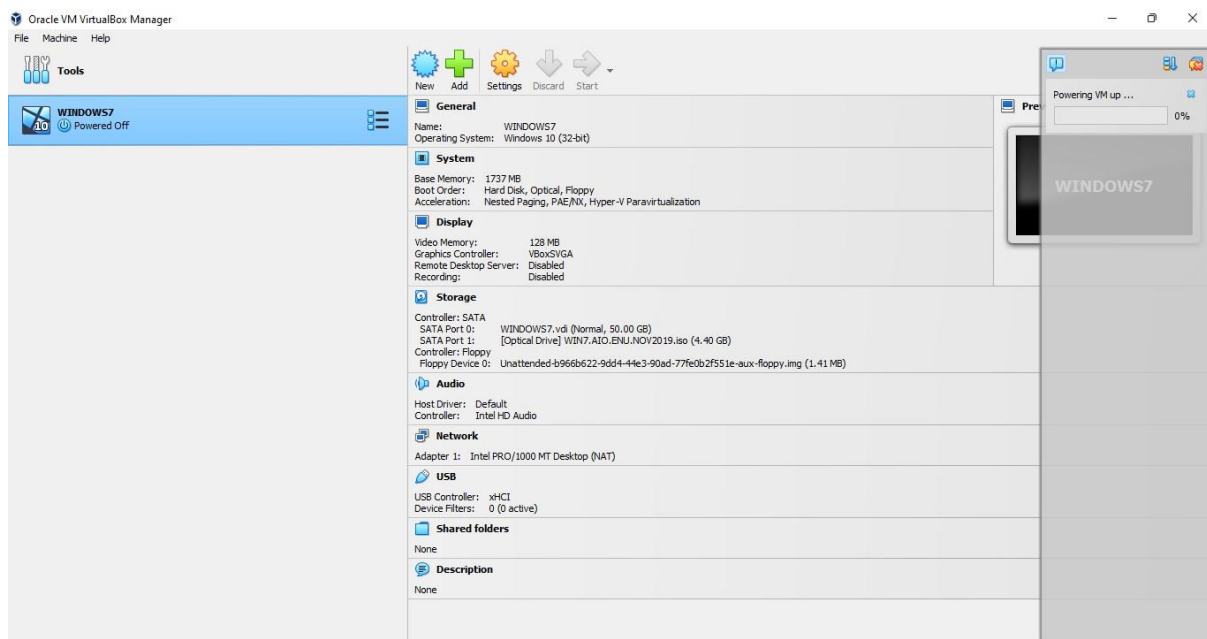
**Step 7: Click on the option to create a Virtual Hard Disk now (should use a minimum of 32 GB for proper functioning). Then click Next.**

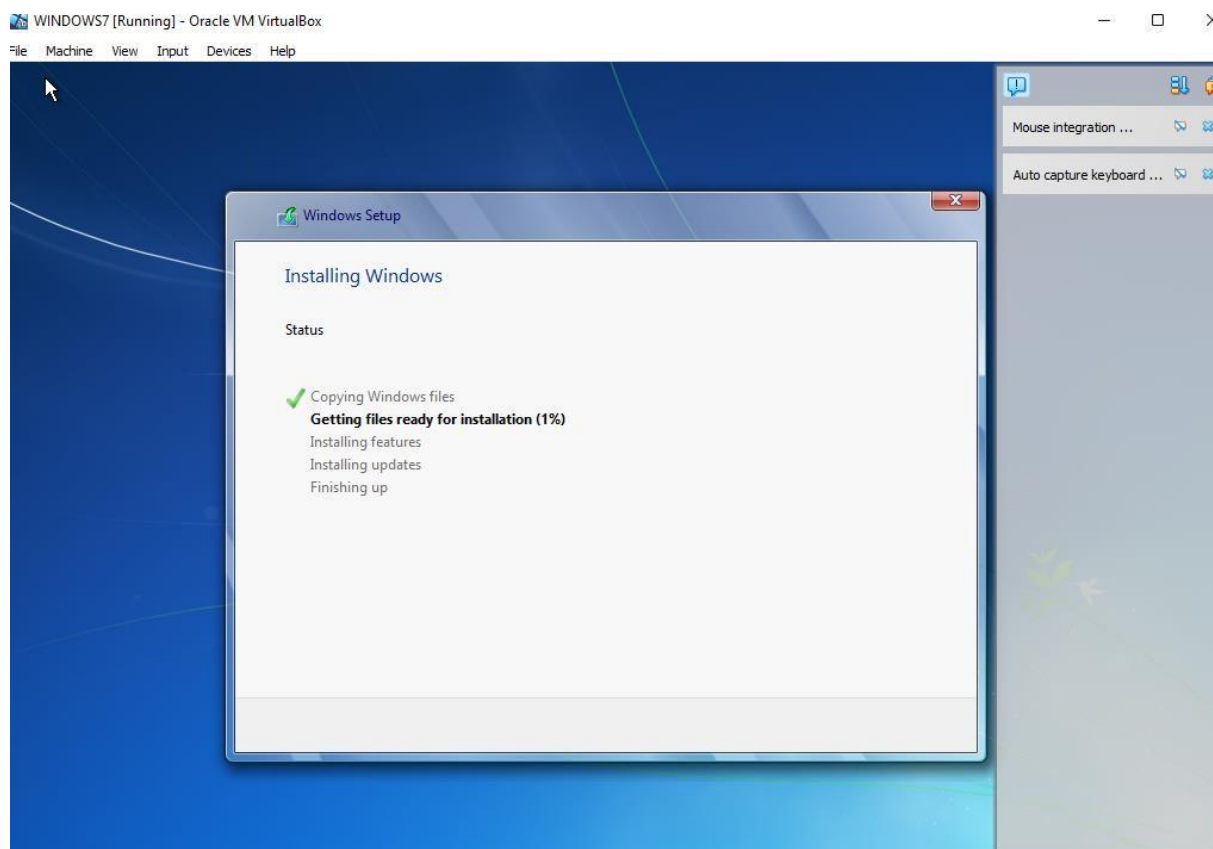
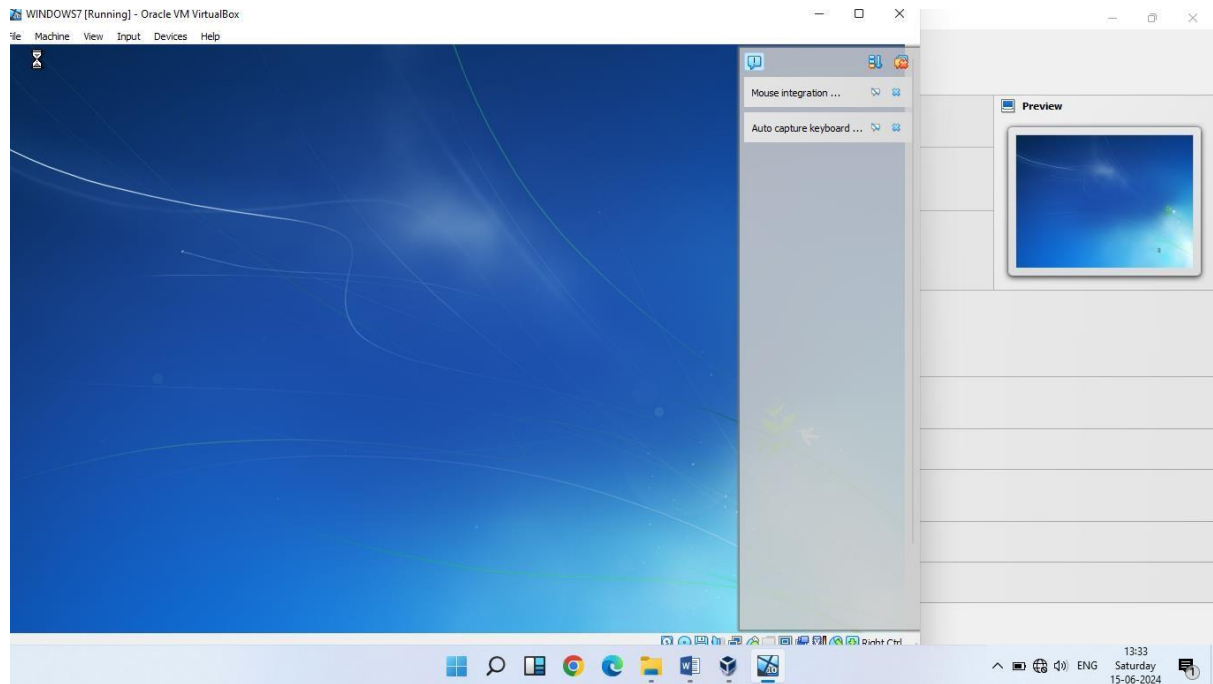


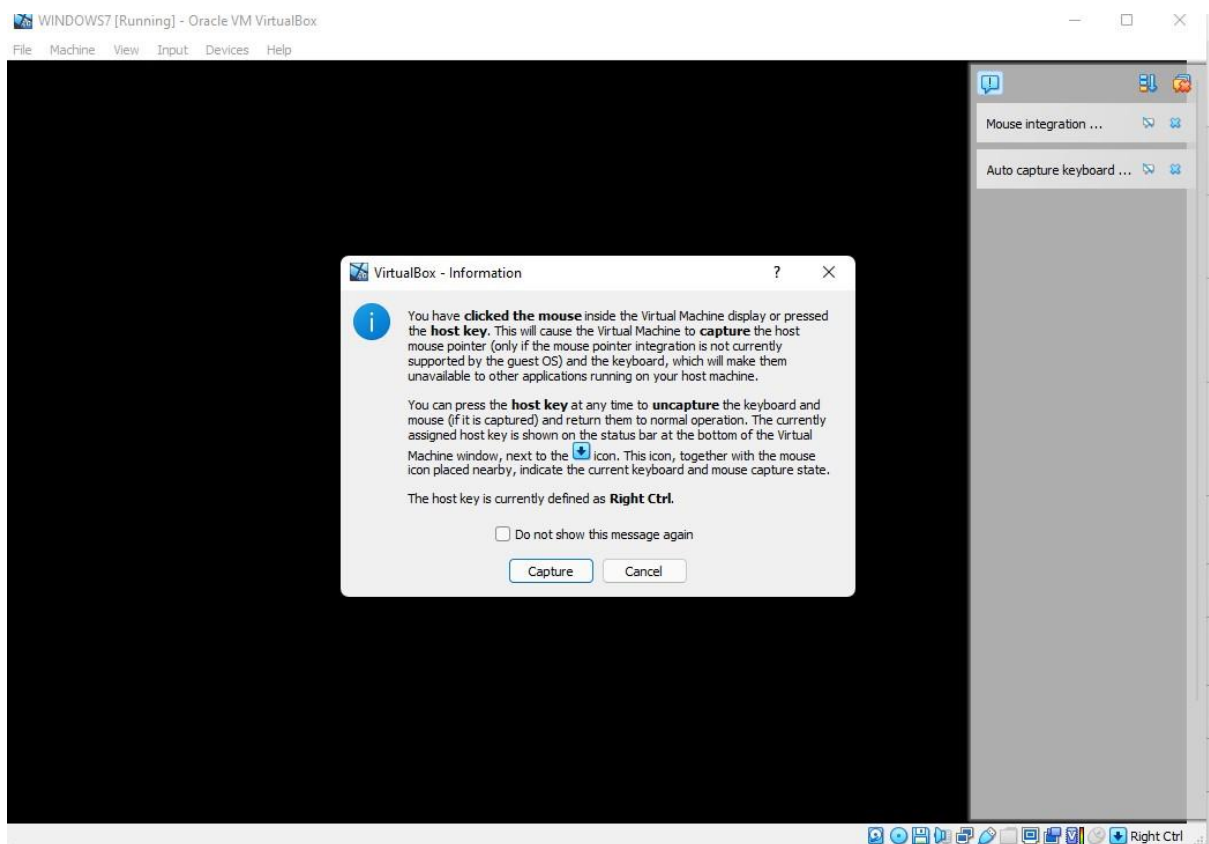
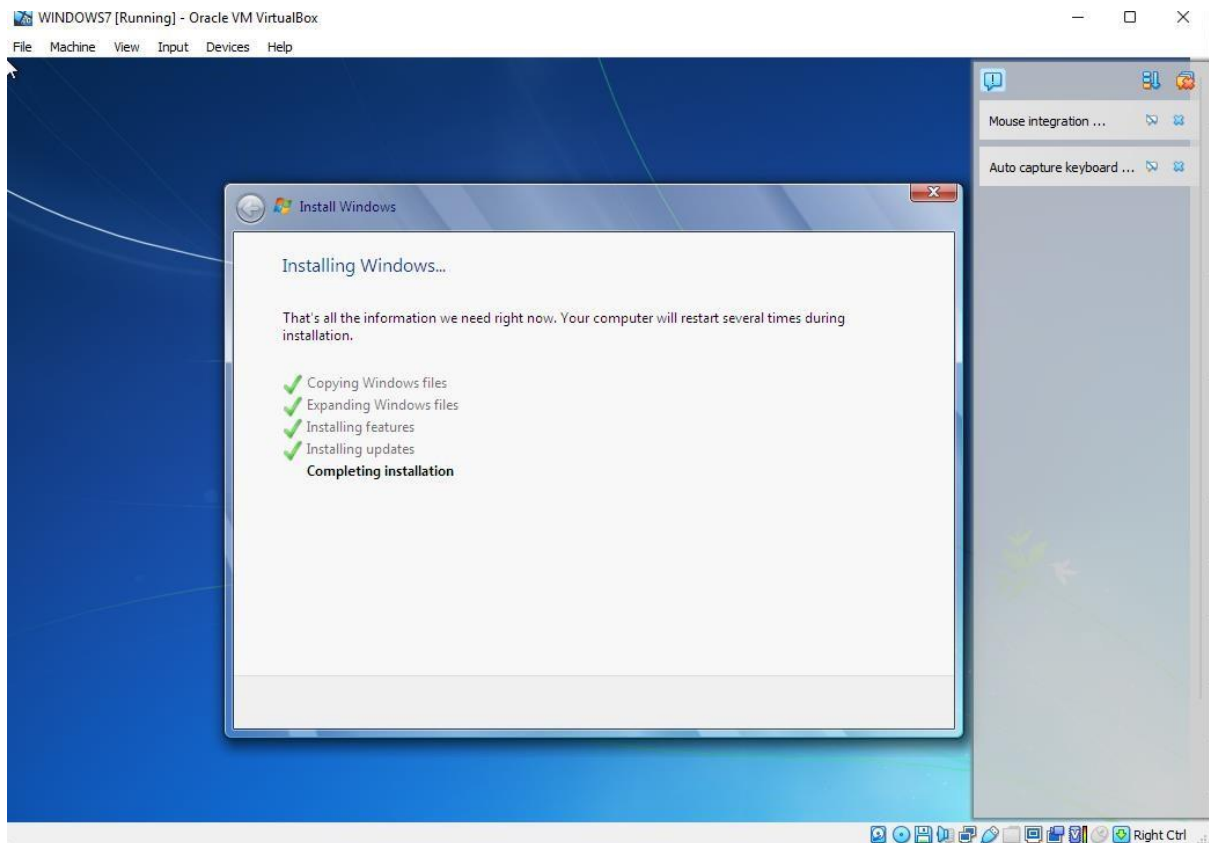
**Step 8: In the following Summary Window, click Finish.**



**Step 9: Click on Finish. After that installation will be started.**



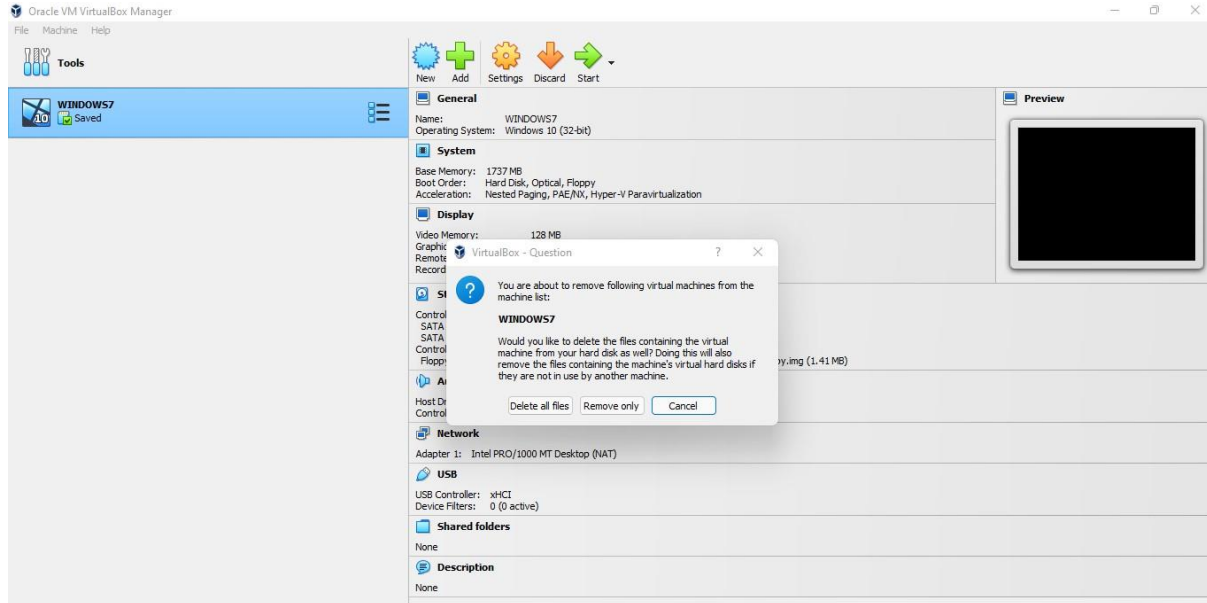




## Step 10: Configure Windows

After the installation, you can configure Windows settings like personalization, network settings, and updates.

Ensure that Windows is activated with a valid license key.



## Experiment: 20

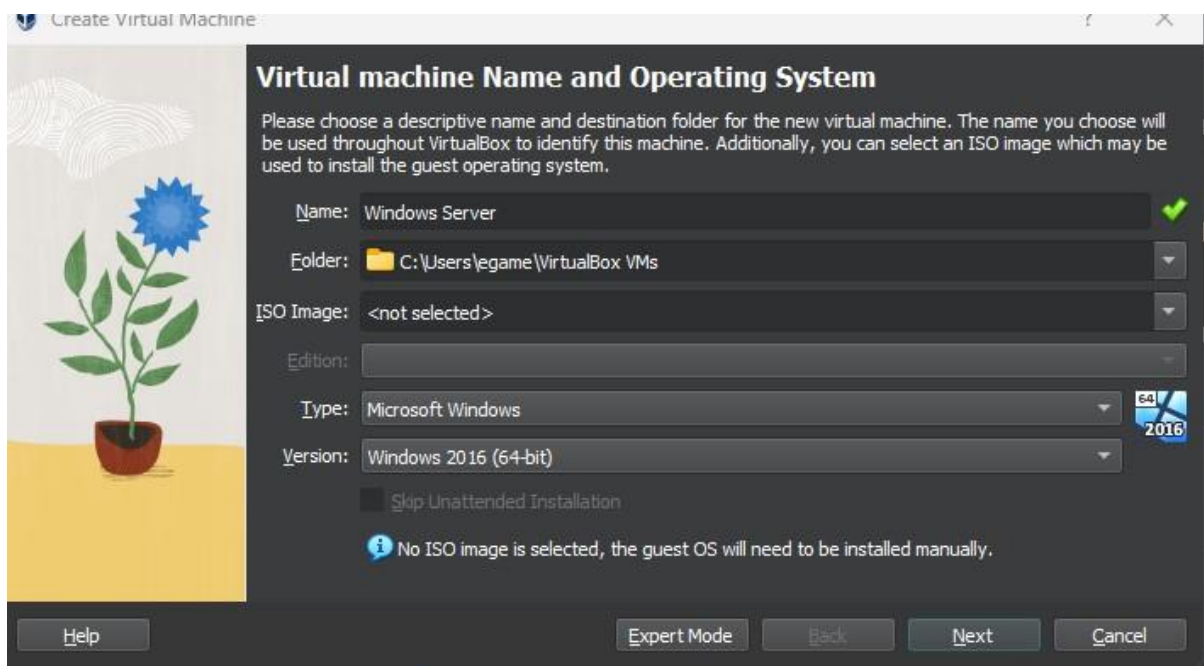
**Aim: Install and Configure Windows Server using Virtual Box.**

**Step 1: Download Virtual Box and Windows Server ISO.**

**Step 2: Install Virtual Box.**

**Step 3: Create a new virtual machine.**

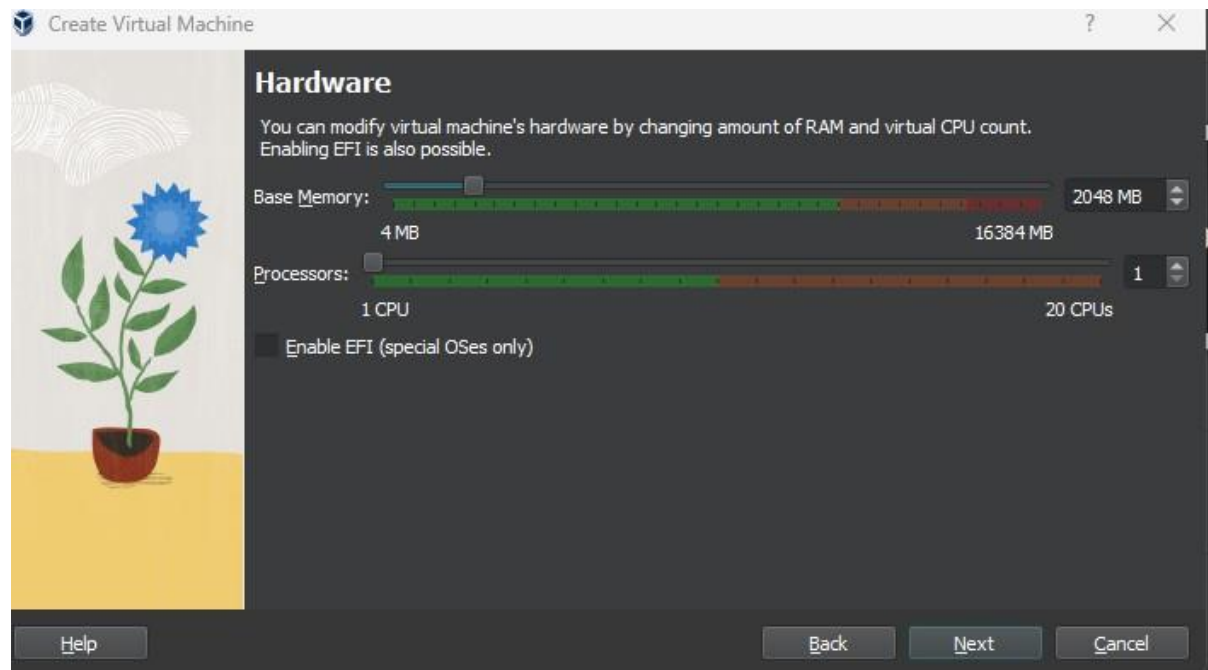
- **Open VirtualBox:**
- Launch VirtualBox and click on **New** to create a new VM.
- **Name and Operating System:**
- Enter a name for your VM (e.g., "Windows Server 2022").
- Choose the **Type** as **Microsoft Windows**.
- Select the **Version** based on the Windows Server edition you are installing (e.g., Windows 2019 (64-bit)).



**Step 4: Allocate Memory (RAM):**

---

Choose the amount of RAM to allocate to the VM. For Windows Server, at least 2GB (2048MB) is recommended, though 4GB or more is better for smoother performance.

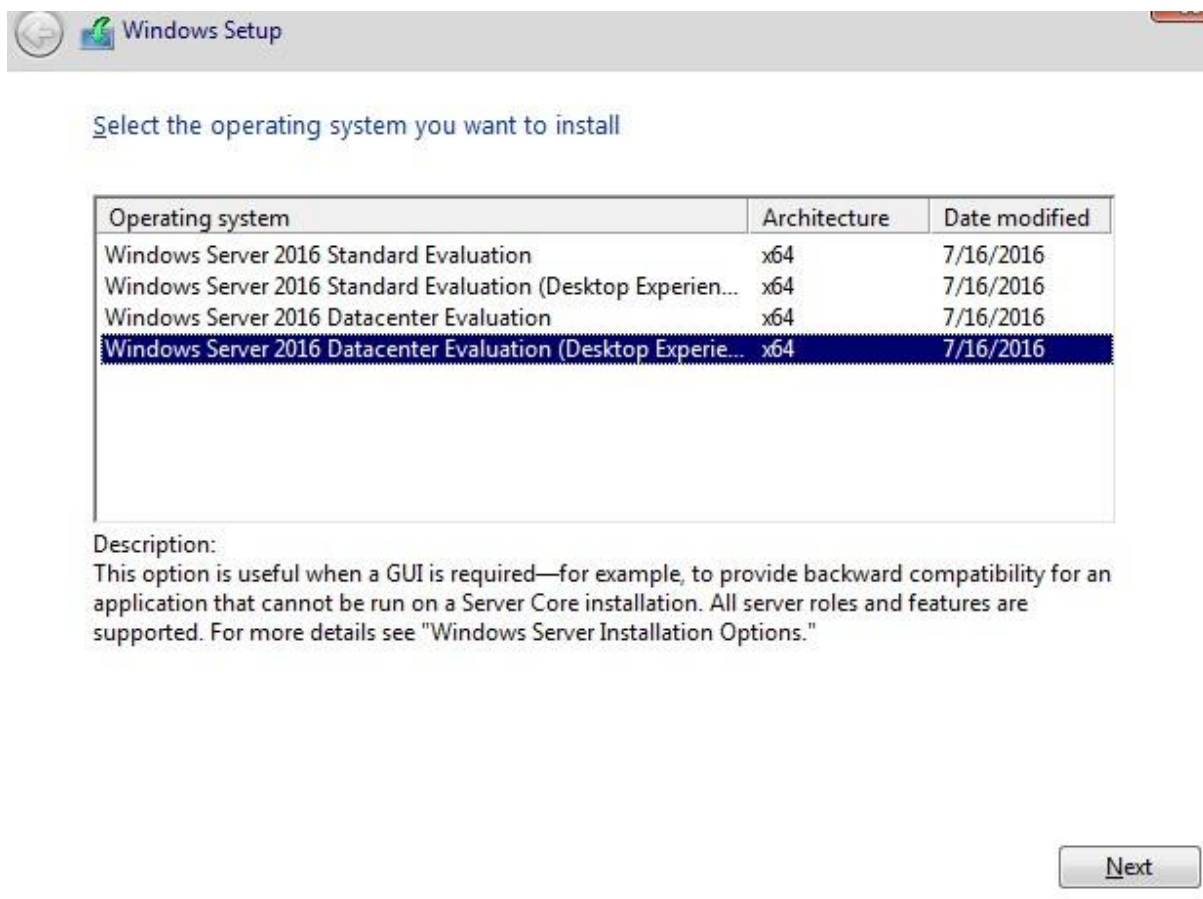


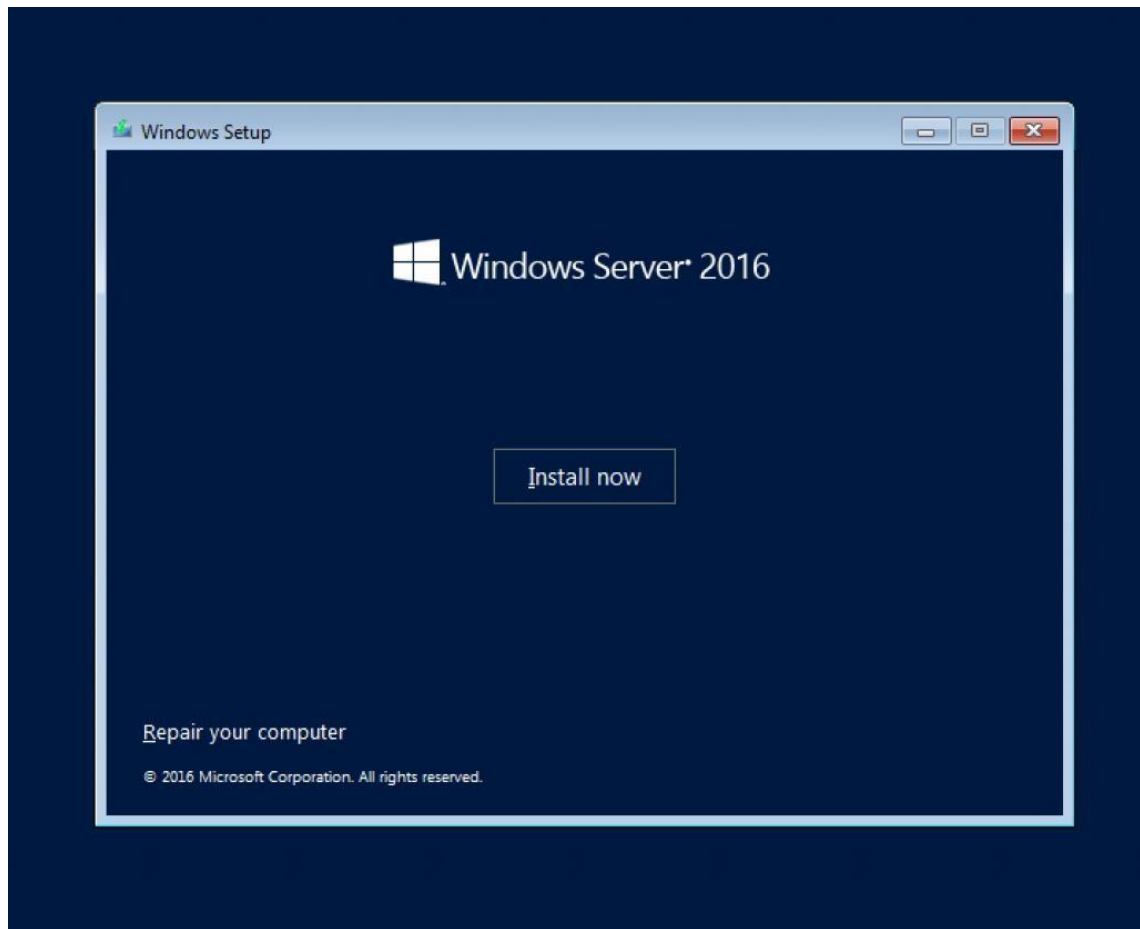
#### Step 5: Create a Virtual Hard Disk:

- Select **Create a virtual hard disk now** and click **Create**.
- Choose **VDI (VirtualBox Disk Image)** as the hard disk type.
- Select **Dynamically allocated** to save space on your physical hard drive.
- Set the disk size (e.g., 40GB).



**Step 6: Click on Install now.**







## Step 7: Post-Installation Configuration

### 1. Install Guest Additions:

- After logging into Windows Server, install the VirtualBox Guest Additions for better performance and features like shared clipboard and drag-and-drop.
- Go to **Devices > Insert Guest Additions CD Image** in the VirtualBox menu.
- Run the installer from within the VM and reboot when finished.

### 2. Activate Windows Server:

- If you have a product key, activate Windows Server through **Settings > Update & Security > Activation**.

### 3. Configure Network:

- Set up the network as needed (e.g., assign a static IP address, configure DNS, etc.).
- You may also need to configure the firewall settings depending on your environment.

### 4. Install Roles and Features:

- Use the **Server Manager** to install roles and features like **Active Directory**, **DNS**, **DHCP**, etc., depending on your needs.
-

**5. Update Windows Server:**

- Run Windows Update to install the latest updates and patches.

**Experiment: 21**

**Aim: Install and Configure Wireshark and demonstrate network traffic analysis.**

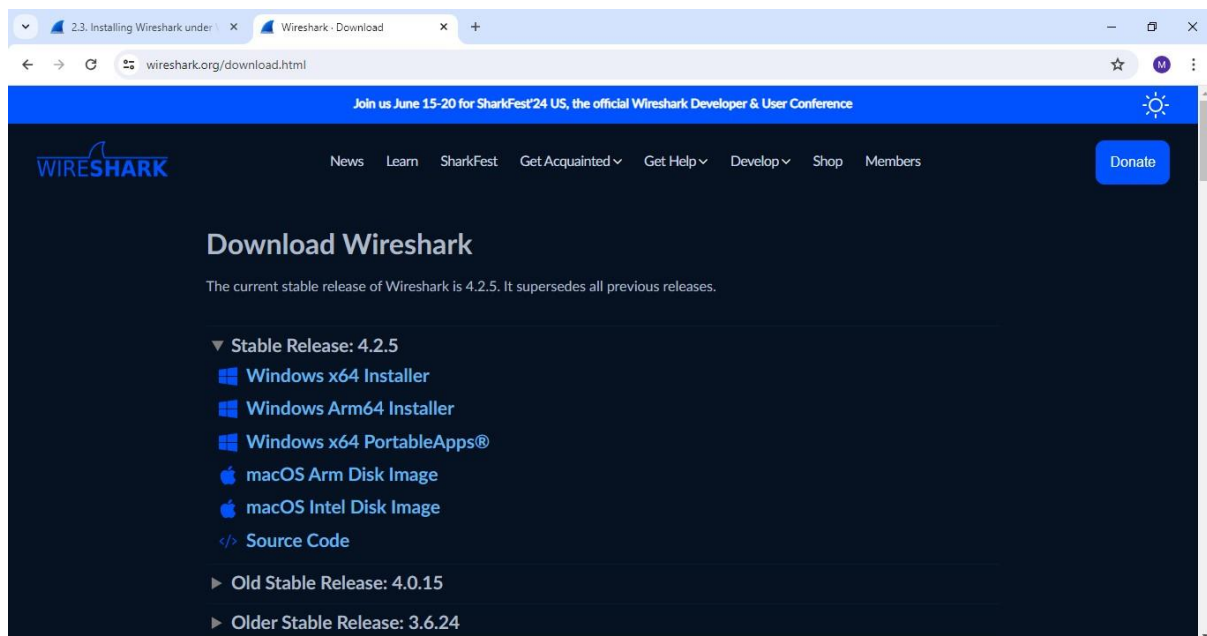
**Introduction:**

---

Wireshark is software that is widely used in the analysis of data packets in a network. Wireshark is completely free and open source. This packet analyzer is used for a variety of purposes like troubleshooting networks, understanding communication between two systems, developing new protocols, etc. The original name of Wireshark was Ethereal which was changed in 2006 due to some company's copyright issues. Wireshark is a cross-platform software, it can be run on Linux, windows, mac, and any other operating system.

### Installation and Configuration of Wireshark on Windows:

**Step 1:** The official Windows packages can be downloaded from the Wireshark main page or the download page. Installer names contain the version and platform. For example, Wireshark4.2.5-x64.exe installs Wireshark 4.2.5 for Windows on 64-bit processors.

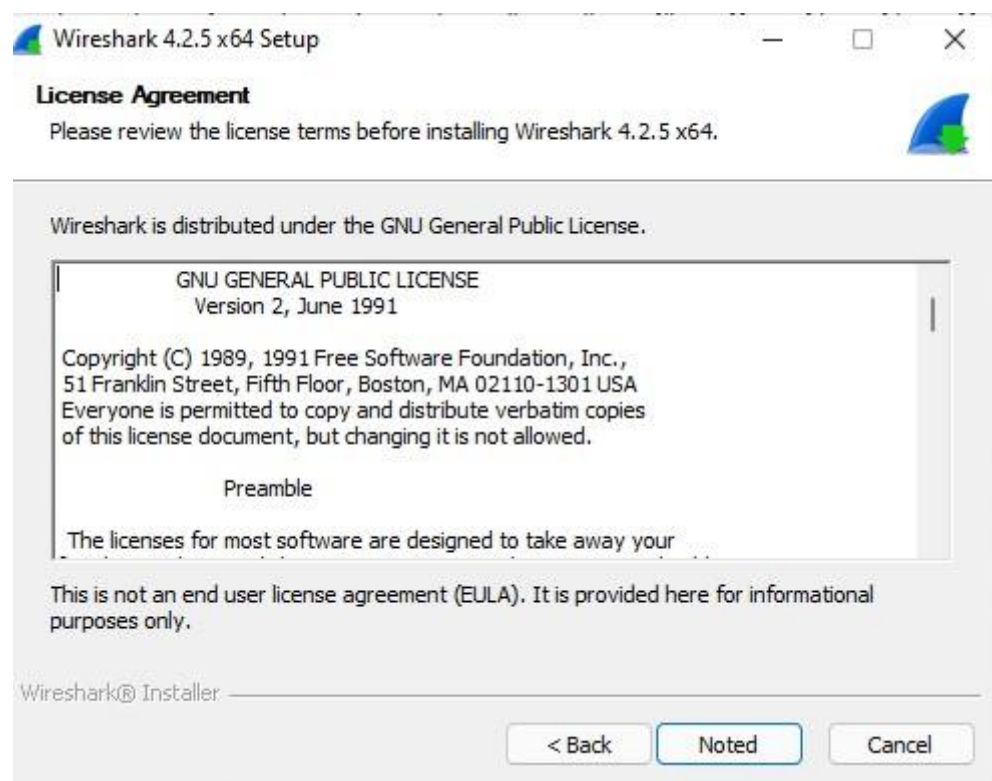


**Step 2:** Check for the executable files in downloads in your system and run it. Setup screen will appear. Click on Next.

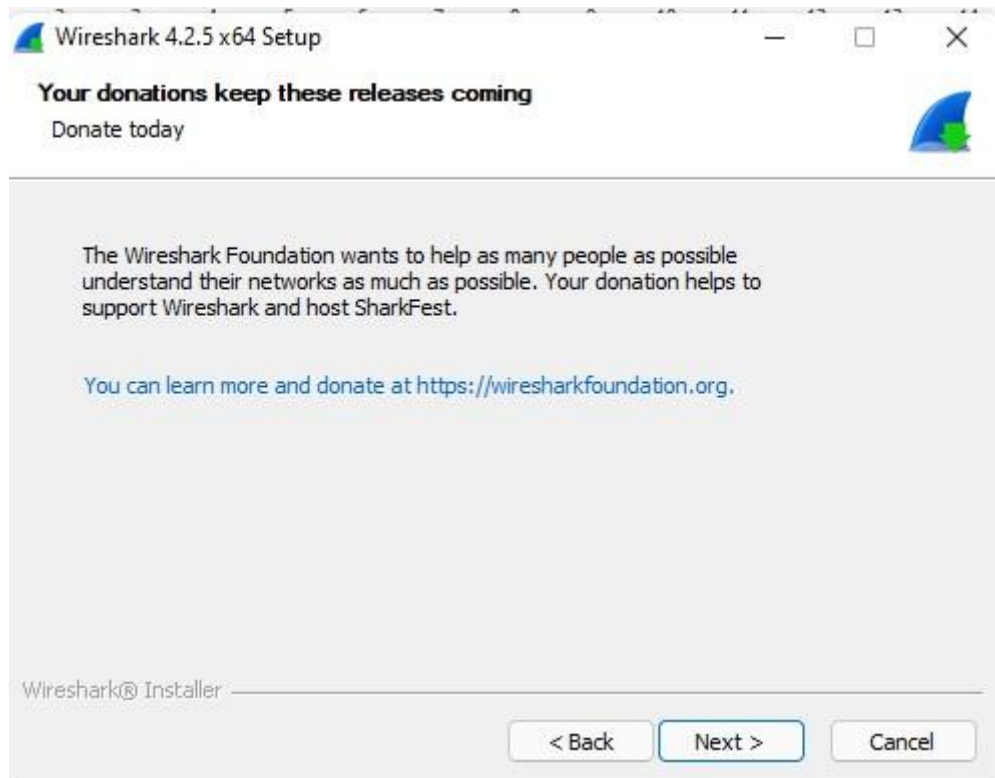
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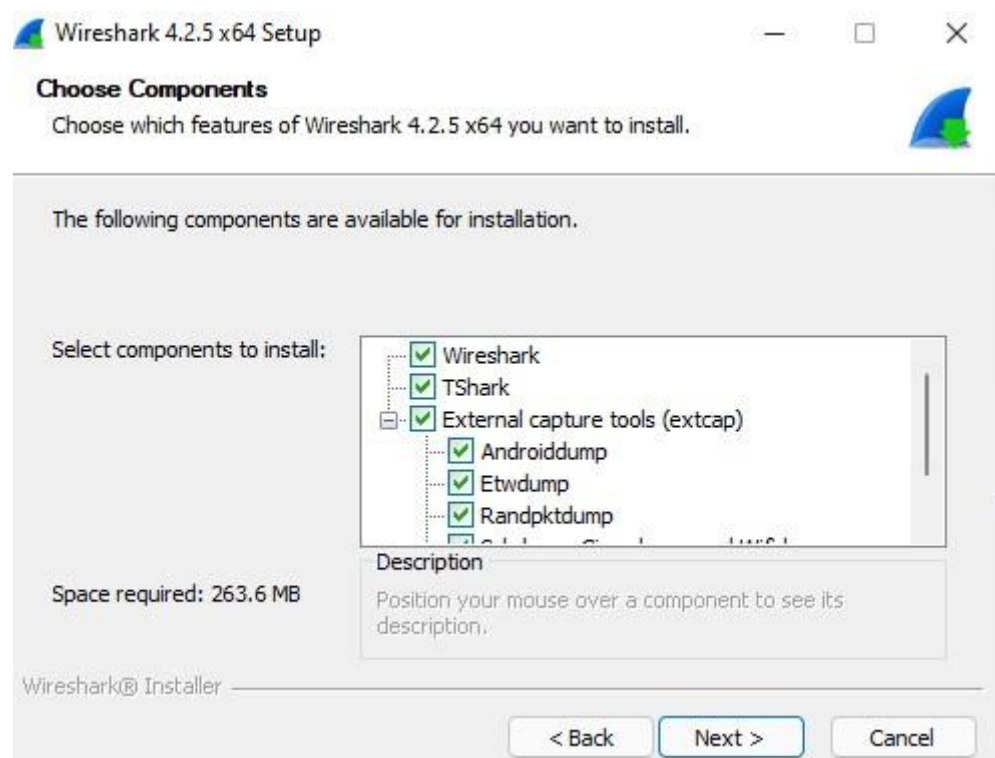
**Step 3: The next screen will be of License Agreement, click on Noted.**



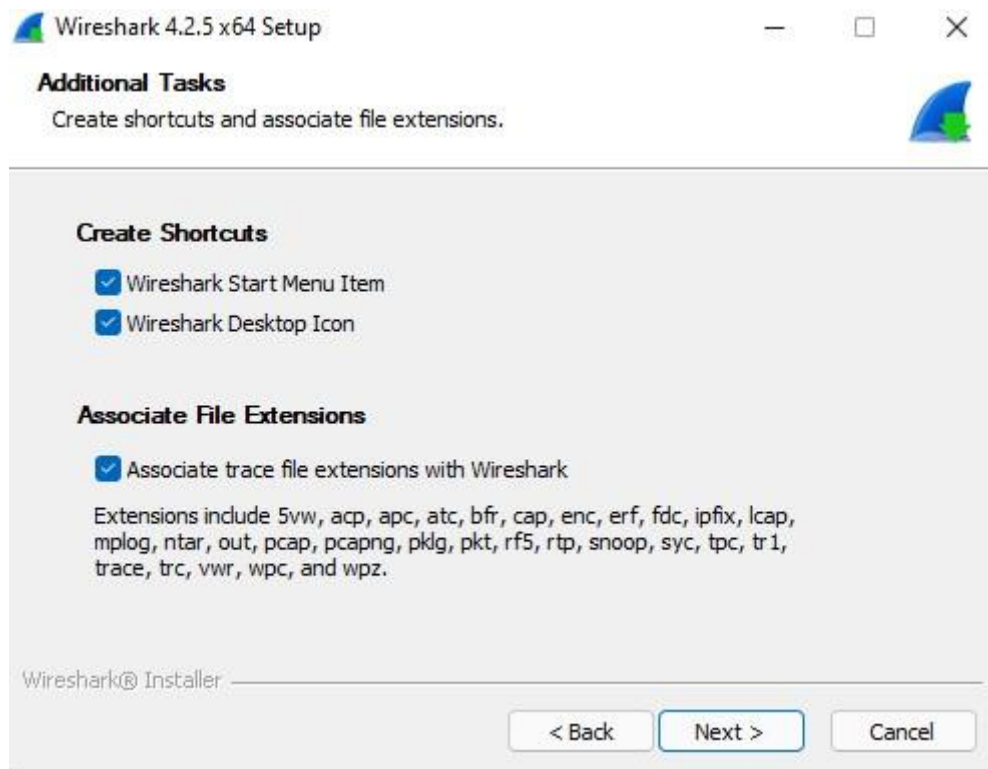
**Step 4: Click Next.**



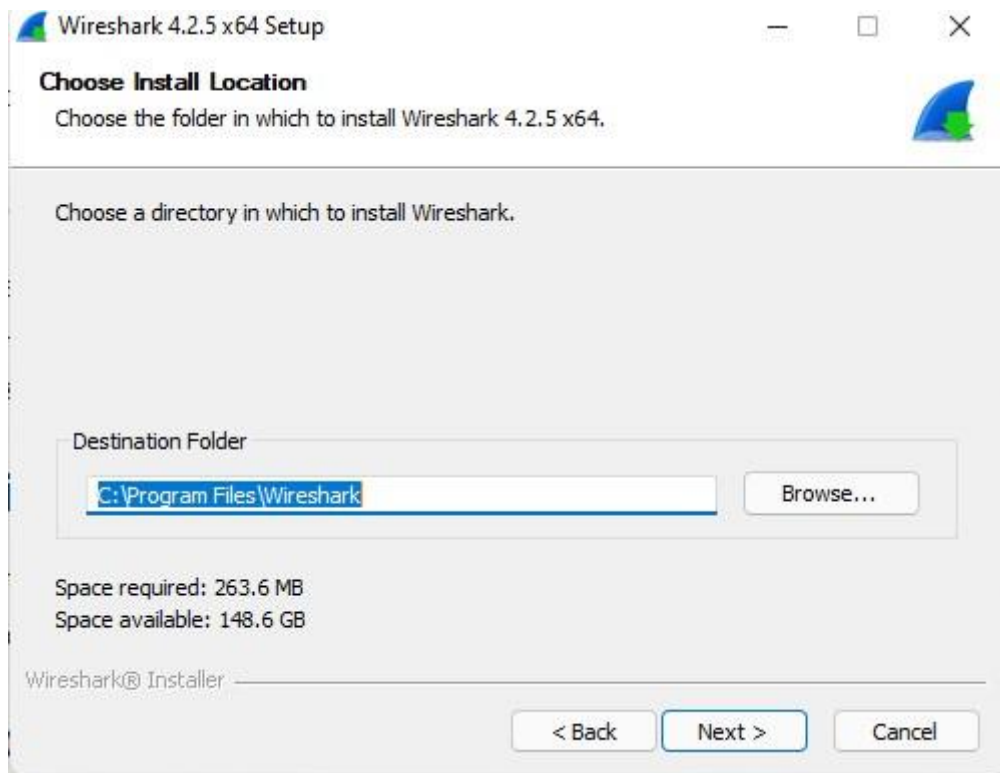
**Step 5: This screen is for choosing components. If all components are already marked then don't change anything just click on the Next button.**



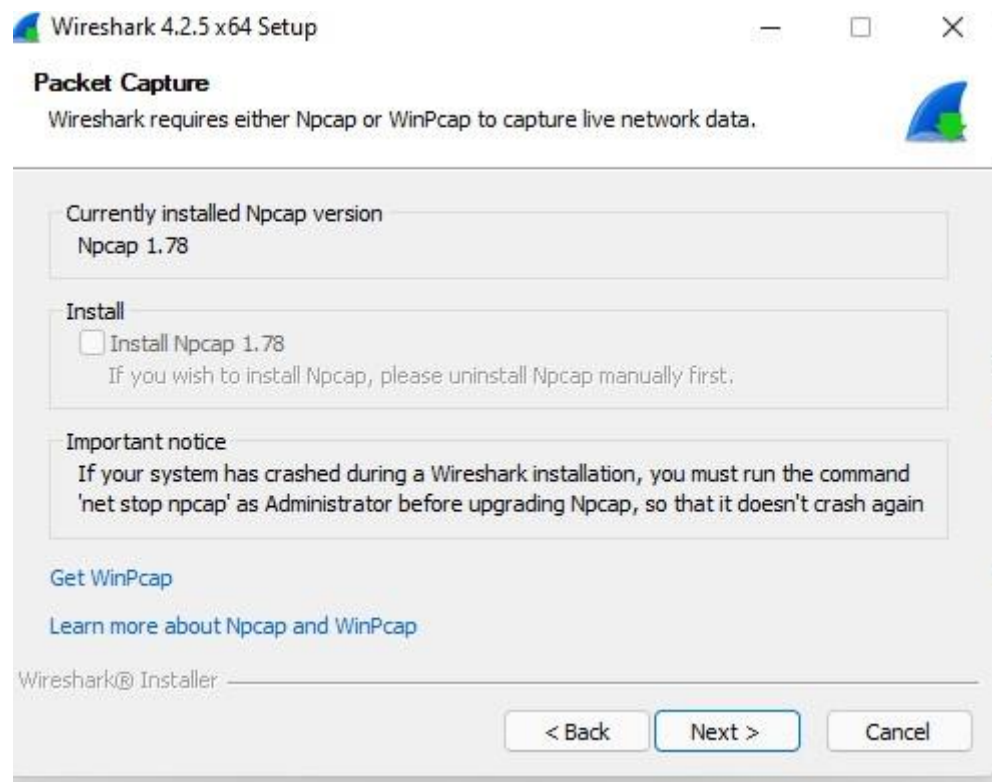
**Step 6: This screen is of choosing shortcuts like start menu or desktop icon along with file extensions which can be intercepted by Wireshark, tick all boxes and click on Next button.**



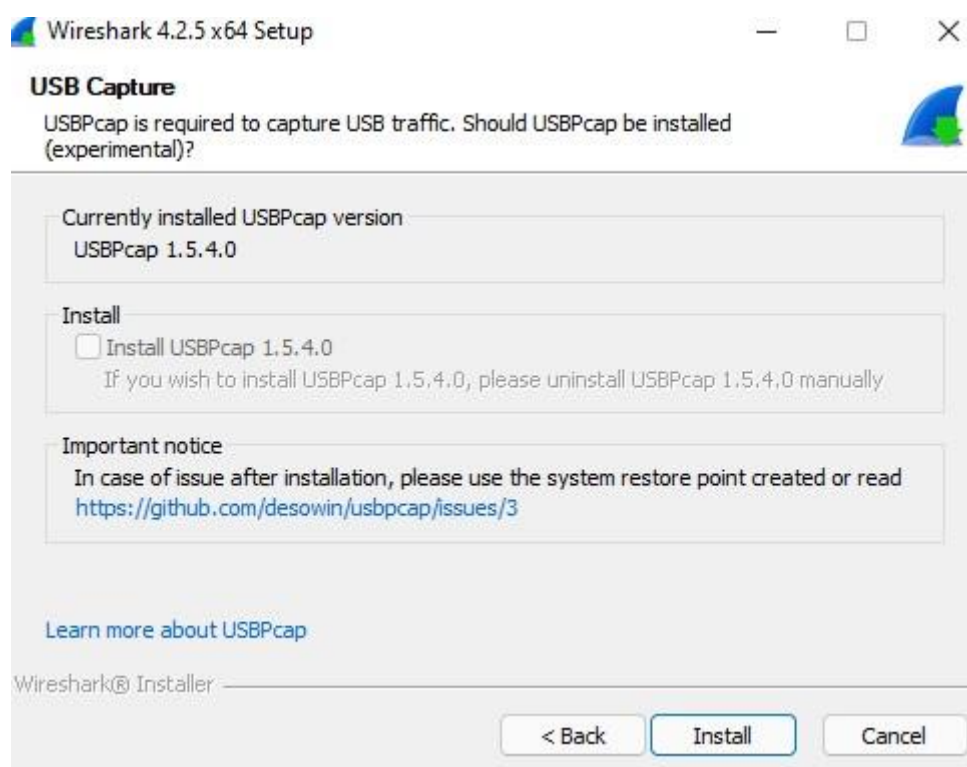
**Step 7: The next screen will be of installing location so choose the drive which will have sufficient memory space for installation. It needed only a memory space of 263.6 MB.**



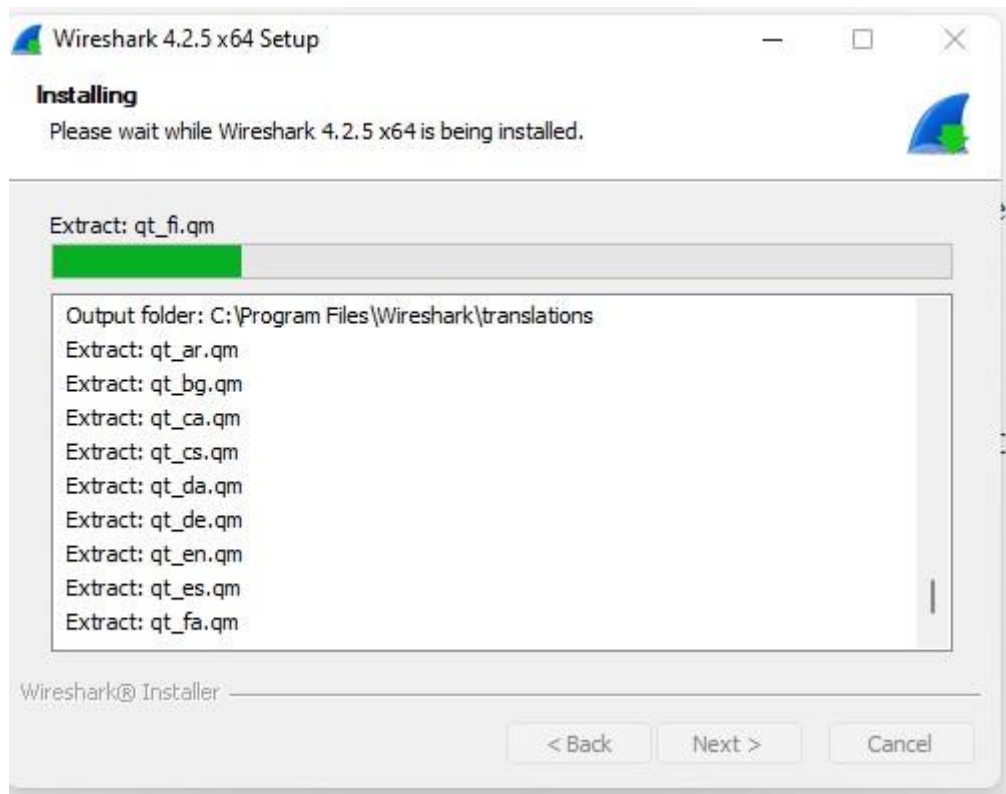
**Step 8: The next screen has an option to install Npcap which is used with Wireshark to capture packets. Click the next button.**



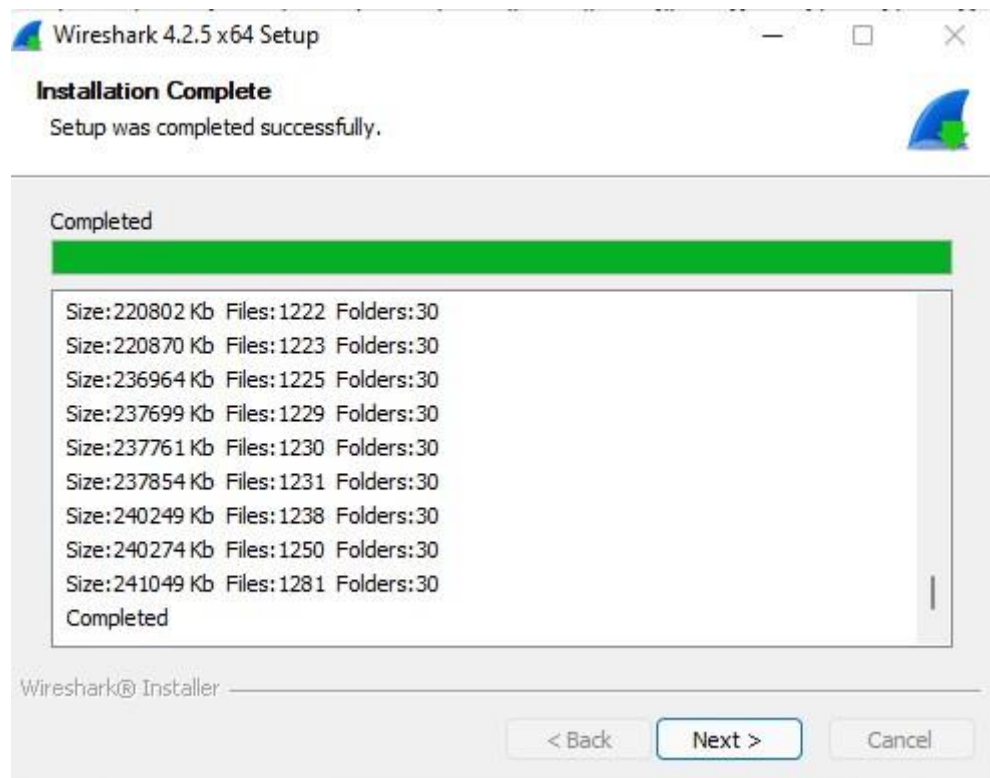
**Step 9: Next screen is about USB network capturing so it is one's choice to use it or not, click on Install.**



**Step 10: After this installation process will start.**



**Step 11: After this installation process of Wireshark will complete click on the Next button.**



**Step 12: Click on Finish after the installation process of Wireshark is complete.**



**Step 13:** Wireshark is successfully installed on the system and an icon is created on the desktop. Now run the software and see the interface. At this point, you have successfully installed Wireshark on your Windows system.

